

BLOCKCHAIN **STARTUPS**

**BITCOIN AND ETHEREUM
AS THE FRONTIER OF FINANCE**



STEFAN KING

BLOCK **STAR**

**BITCOIN AND
AS THE FRONT**

This book shows you how to compete with banks. Bitcoin and Ethereum have features that empower entrepreneurs to disrupt finance. The functions of banks are now in the cross-hairs of open-source groups and software entrepreneurs. We look at what you can build for the future beyond traditional banking.

Banks used to be protected from disruption by the nature of fiat money. The lack of competitors has made them slow to innovate. With Bitcoin and Ethereum, software entrepreneurs can spot opportunities to offer state-of-the-art financial services and soar above the status quo.

BLOCKCHAIN STARTUPS

Bitcoin and Ethereum as the Frontier of Finance

STEFAN KING

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To the crypto-wizards of Chiang Mai.

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Acknowledgments

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Introduction

ONCE A SOFTWARE DEVELOPER asked me about the possibilities of Bitcoin. I explained it like this:

Me: ‘Have you done work for clients as a freelancer?’

Developer: ‘Yes.’

Me: ‘Then you are familiar with the challenges that come with independent contract work; you have to make deals such as getting 50% payment upfront and 50% after completion of the project. Or you have to use an escrow service. You need contracts, and you and the client take on risk. Sometimes it gets messy, and you could have conflicts about payment.’

He nods quickly; he knows how things go. I continue: ‘People are currently working on technology that works on top of Bitcoin and other cryptocurrencies that solves this, and many other things. For example, you could define a smart contract that pays according to your project’s timeline. Say that you and the client expect it to take two months to complete and that the price is fixed. The smart contract starts the money stream from the client to the contractor, minute by minute, until the end date. If someone is not happy at any point, either you or them clicks the button, and the stream of money stops. This guarantee is unstoppable and does not require an escrow agent.’

The developer understands it, because he says: ‘I’m going to work on this as soon as I get home.’

Such moments of entrepreneurial inspiration are typical for people who study the details of cryptocurrency.

This book aims to help entrepreneurs make sense of the opportunities offered by cryptocurrencies and digital tokens. The launch of Bitcoin opened the field of finance for disruptive innovation: new products and services are possible, built by up-and-coming players such as startups and open source projects. In the age of crony-capitalism, which abuses its political power to earn money without risk, a handful of companies control finance. Other problems include fraud, quantitative easing, and too much

debt. That is why innovation in the financial sector is an exciting trend. The invention of Bitcoin and decentralized finance means that you, yes *you*, can dive into the field of financial technology and stake a claim on the profits of banks. You can start solving problems that legacy players are ignoring.

The big-sounding phrase “global financial sector” suggests that a scrappy entrepreneur like yourself should stay away from it. Any financial product, such as an insurance policy, mortgage, or investment, involves the creation and transfer of money, which are critical functions in the economy. That is why the “global financial sector” runs on top of banks. We can even be more specific: It is an oligopoly of institutions that controls money and investing (Andreas M. Antonopoulos, *The internet of Money*, 2016). Economics 101 says that an oligopoly is a group of large companies within an industry who agree to keep prices high and competition at bay. Big banks work together to achieve a *de facto* monopoly on financial services. The cronies fear startup companies that compete with them on a level playing field. Your challenge as a crypto entrepreneur is to bypass and undercut these banks by giving their current customers a better financial product or service. You can win by inventing value propositions that outclass those offered by banks.

This book is for doers and builders. I will focus on your opportunities as an entrepreneur. That is why we will skip cryptographic math, computer science, economic models, or game theory. Of course, these topics do matter: they are part of your arsenal as an entrepreneur, and it is to your advantage to learn as much about them as you can. But these topics have already been covered in other books, which I will mention in this book. Whenever I reference a book about cryptocurrency and related issues, it means that I recommend it with enthusiasm.

The history of technology has shown how disruptions play out. Innovators build a scrappy application or platform that grows from obscurity, powered by a new perspective on an existing product. They get rapid feedback from their initial user base, taking advantage of their greater adaptability as a small organization until they boost the user experience beyond those of the legacy players. This process causes drama. Every technology that makes our life more efficient replaces companies that served that market less efficiently. For example, the invention of clay roof tiles put artisans of thatched roofs out of work, and the mainstream adoption of cars turned horse-drawn carts and carriages into quaint antiques.

Likewise, Xerox machines and personal computers replaced armies of secretaries. We often see that technological progress gives a boost to the quality of life of the many at the cost of a forced career change for the few. Recent examples relevant for software entrepreneurs include brick-and-mortar retailers who lost their business to e-commerce platforms. It is easy to see how we all benefit from progress. If you have lived through any of the above examples, you know that feeling you get as a customer: “Wow! This is so much better! From now on, I want this new thing.” Cryptocurrencies and other crypto-assets can be like that. Entrepreneurs can use blockchains such as Bitcoin and Ethereum to disrupt banks, who are the legacy players of finance. Cryptocurrencies put the control over money back into the hands of the public.

Economic progress depends on entrepreneurs applying technology to people’s problems, and most of us have plenty of financial challenges. Society, as a whole, has large-scale money problems that remain unsolved. A startup can tackle any one of these problems, which we will discuss one by one. By dividing and conquering, we can transcend the widespread economic and political challenges related to money and finance. Entrepreneurs have a heroic role to play; taking honorable risks to get rewarded for mastering these challenges. Cryptocurrencies are the tools to get it done.

The inventors of Bitcoin and Ethereum designed them for smooth financial transactions. In contrast, banks have an outdated infrastructure with friction and hidden risk in most transactions. The entrepreneurs who drill into these challenges have an excellent chance to succeed with one venture or another. Their strategy is simple: 1) look at financial functions that make customers vulnerable to a third party, that is, someone other than the sender and receiver of an asset, and 2) automate that function on a public blockchain, which acts as an immutable ledger of transactions. International transfers of a cryptocurrency are already superior to the services of any bank or other financial institution. You do not even need to have an account anywhere. Furthermore, software entrepreneurs and finance wizards can build services that use this money without having to partner with a bank. Chapters 4 and 5 explore some financial solutions that don’t rely on the legacy banking system. Chapter 6 gives examples of successful tokenized assets.

As I write this in 2020, only a small portion of the global population wants cryptocurrency. However, when Bitcoin and Ethereum work as intended, traditional financial products will become obsolete. In the future, other blockchain-based systems could become relevant as well. Where a public blockchain can replace intermediaries, transactions get faster while costs fall. For now, the use cases are limited to international payments, some illegal trades, unstoppable donations, and protection from inflation. We are still figuring out how to apply this technology to other commercial uses.

From the broad adoption of paper currency until the invention of Bitcoin, the business of creating currency belonged to banks. They profit from low interest rates on loans from the central bank, they create new money, and they get exclusive access to the best investment opportunities. In their comfortable status quo, banks capture most of the value of their captive customer base. However, since the launch of Bitcoin, the power of banks is up for liquidation. Nowadays, entrepreneurs can compete for customers by using blockchains such as Bitcoin and Ethereum. A superior class of value propositions is now possible. We can enter existing but closed markets and discover new ones, for example, serving people who do not have bank accounts, the so-called “unbanked,” or explore new business models for selling digital content. We can develop products that benefit from the shared efforts behind open-source software. The entrepreneurs and their customers take the power from banks by transacting on a new network that holds cryptographically secured assets.

Entrepreneurs bring crypto-assets to the market to solve people’s financial problems. The energetic founders who see the future as open and decentralized have the right vision for this new technology: disruptive money. They understand the promise, and they are itching to build. This book is for them.

Starting a company is an adventure because the blockchain industry resembles the Wild West. As you travel in the unknown terrain of Bitcoin and crypto-assets, you are going to want a decent map. Some can sell it to you. But who can you trust? You need some way of navigating this new space. You can get help from someone who has been out there already, who got his hands dirty and is familiar with the territory. This book is the “map” for an entrepreneurial venture that makes use of Bitcoin and Ethereum or any other blockchain flexible and secure enough for the service you plan to

build. I based it on my hands-on experience working with blockchain entrepreneurs, developers, and consultants. As I was not founding an organization myself, I had time to observe and reflect, to make sense of the chaotic blockchain hype. I maintained perspective, and here I attempt to share my view with you in a way that I hope is useful to your entrepreneurial challenge. I worked for technology startups for over a decade as a consultant, copywriter, and project manager. I marketed high-tech value propositions. Since June 2017, I applied that skillset in startups that issued and traded crypto-assets. I have worked in teams that dealt with the challenges of designing a token economy and crafting a compelling value proposition for their users. I connected with entrepreneurs, developers, and marketers who showed me their projects. I have seen a lot of brilliance, but also a lot of window dressing and cheap talk. Throughout the blockchain hype of 2017, I was immune to the aggressive marketing campaigns because I studied the fundamentals of Bitcoin, and to a lesser extent, the workings of several other coins. I started following the industry in early 2013 and began building relationships with crypto developers and entrepreneurs since 2014, people who were working on the first wave of alternative coins. I got involved more seriously in 2016, inspired by the weekly Bitcoin meetup in Chiang Mai, Thailand. My priority was to make sure that I knew how Bitcoin worked in detail. Understanding the nuts and bolts of Bitcoin as well as its historical background protected me from being hypnotized by mirages or bamboozled by buzzwords.

I want your venture to succeed because people like you will make a difference for the next decade of financial technology. The megatrend of decentralization, which gained speed with the internet, is moving into third gear with Bitcoin. I am excited that this ecosystem of exponential growth can take you and your venture to success.

The best practices for blockchain ventures have their foundation in existing startup lore. We have the codified knowledge about starting a technology business with an idea and validating that idea by letting the market respond to your product. We have scripts for pivoting and scaling up. Now, we need to add an understanding of blockchains to that tradition and see how we open a new marketplace. Beyond a few certainties regarding Bitcoin and other blockchains, the future development of this technology is unknown. We will address that unknown by relying on rules of thumb, and we can walk through historical trends to find parallels.

The book is organized like this:

Chapter one, *Disruptive Money* , examines the role of money in the economy. To act as effective disruptors, entrepreneurs need to know their history. Various types of money have different origins, and people use them for different reasons. You can compare Bitcoin to traditional money, in terms of its features, advantages, and development.

Chapter two, *The Political Entrepreneur* , shows how Bitcoin empowers a new kind of startup. Disruptive money systems, such as Bitcoin and Ethereum, define a territory of freedom, new virtual jurisdictions that protect the assets living on them with new security features. We can compare the relative security of blockchains and their resistance to the authority of governments.

In chapter three, *Liquidity and Adoption* , we see that people adopt crypto-assets that fulfill a need. Crypto-assets with a commercial function catalyze a network effect. Rising liquidity makes the asset more stable, which draws new users, making the asset more liquid. A crypto startup has to place itself in that feedback loop.

In Chapter four, *Ideal Money* , we see how a layer of secondary protocols can make money as efficient as information. The currency will free itself from the specific media that now dictate how it can move. Transacting with minimal friction and smart contracts could unlock new types of business.

Chapter five, *Wallet Design* , is about removing any friction when handling money. The mainstream will only adopt crypto-assets that are easy to use; such a design is a creative challenge. Various hardware and software approaches make tradeoffs between convenience and security.

Chapter six, *Tokenized Assets* , asks: Besides money, what else can we put on a blockchain? Bitcoin sets a standard for freedom and security to which other blockchains can aspire. Teams that aim to expand the possible use cases make designs for a new economy. These tokenized assets promise a future of trade and prosperity for the mainstream, provided that designers solve incentives, power, and governance.

I assume that you sense the opportunities afforded by the proven success of well-known blockchains such as Bitcoin and Ethereum and that you are adventurous. You can handle a period without the certainties of proven customs and established laws. You are an explorer with the guts to

stake out a territory of freedom. If this describes you, welcome to the Wild West. Here is your map. Enjoy the journey.

CHAPTER 1

Disruptive Money

THE FINANCIAL INNOVATION that is Bitcoin is still young. At the time of writing in 2020, Bitcoin is, for the most part, a niche product for specialized developers and speculators. Only a small portion of the world population owns crypto-assets, and even fewer understand how to hold and manage these assets. The prices follow hype cycles, driven by hope for the future and a rollercoaster of news, which makes it an environment of pioneers. The challenge is to deal with the confusion. Those who pierce the fog and see clearly, have a chance to seize opportunities as innovators.

Startups have a chance to innovate when their team builds a product or service with the tools of an established blockchain, such as Bitcoin or Ethereum. The benefit of either of these networks is their solid history, which enables your company to offer a reliable, uncensorable transfer of digital assets, becoming part of your competitive advantage when offering a new financial product or service. Smarter transactions for challenging situations are the future of money and finance because this is the latest industry to be disrupted by software. Software innovation took a long time to come to money because the intellectual challenge of inventing digital scarcity was hard to surmount. We can measure the size of that achievement by the political power required to create scarcity in ordinary paper currency: only an entity as large as a nation-state has the clout to keep a national currency secure. Furthermore, the technical work on cryptocurrency is a work in progress. As it is, functional national currencies have little to prove because the institutions of that currency are the standard for payments and investments. We need to build on top of the insights of Bitcoin to let the internet support money and finance. Bitcoin and Ethereum are the first layer of private money, the first challenge to banks. We have to apply these insights to a growing set of value propositions. Blockchain products and services are not automatically better than those of banks. They merely have the potential to be so. The fight for innovation has only just started.

We will use these definitions from public sources.

According to the Bitcoin wiki:

Bitcoin is a decentralized digital currency that enables instant payments to anyone, anywhere in the world. Bitcoin uses peer-to-peer technology to operate with no central authority: transaction management and money issuance are carried out collectively by the network. (...) Building upon the notion that money is any object, or any sort of record, accepted as payment for goods and services and repayment of debts in a given country or socio-economic context, Bitcoin is designed around the idea of using cryptography to control the creation and transfer of money, rather than relying on central authorities. Bitcoins have all the desirable properties of a money-like good. They are portable, durable, divisible, recognizable, fungible, scarce, and difficult to counterfeit.

According to Ethereum.org:

Ethereum is a global, open-source platform for decentralized applications. On Ethereum, you can write code that controls digital value, runs exactly as programmed, and is accessible anywhere in the world.

According to the Bitcoin wiki:

Blockchain is a cryptographically secured, time-stamped, public and distributed database of every (...) transaction that has ever occurred on the network. “Distributed” here means that the information in the blockchain is broadcast to and recorded by every node in the network. There is no one central database.

Bitcoin and Ethereum work on similar principles of decentralized transfer of value, but make different choices for their network. Whereas Bitcoin aspires to be a secure and reliable alternative financial system, Ethereum has a broader mission: decentralized software applications, with a higher risk for the assets on its blockchain. Security and flexibility are a fundamental tradeoff, and each blockchain chooses a balance between them, consciously or not.

Within the world of finance, Bitcoin, and to some extent Ethereum, solve the problem of centralized control over transactions. Bitcoin and Ethereum empower their users by liberating them from any controlling or regulating third parties. If cryptocurrencies are a new asset class, then the transactions they enable are also of a new class: digital transfer and value-holding, without a third party. Users need a new breed of companies that make those transactions quick, safe, and easy. In the introduction, I described the essence of crypto entrepreneurship as replacing a financial product or service that makes the customer vulnerable to a *third party* : a person, company, or institution that controls the asset besides the parties that transact. Why is the third party a problem? To understand the original reasons for a third party, we have to go over a brief history of finance. We will look at the economic function of each player in financial transactions, to understand the possibilities offered by an established blockchain.

Money as Scarce Stuff

In agricultural societies before the industrial revolution, the money was gold, silver, and copper. At the level of society, money is a way to keep score. Or: Money is a way to keep track of who owes what to whom, and a way to transfer value. That was one of the jobs done by bankers: transmitting payments. One of the essential functions of banks is to move money from one place to the next while keeping it secure. Bankers settled the balance of trade between cities and countries by transporting gold and silver, equal to the difference of the value of traded goods. For this, they invented cheques, also known as *bills of exchange* . Think about the age in which European countries connected, with each other and other continents, in the 16th or 17th century. Goods, such as spices, flowed from Amsterdam to Brussels, and were paid for with an equivalent flow of silver from Brussels to Amsterdam. At the same time, there was a reverse flow of goods, such as wine, from Brussels to Amsterdam, and silver going from Amsterdam to Brussels to pay for that wine. Silver had to be carried around, which had a cost and a risk. That meant that transporting money was costly; it was, after all, a precious, physical commodity, and at risk for robbery. Bankers made most of these physical transports unnecessary by issuing cheques to their clients, which they could cash in at the corresponding bank in another city. So instead of moving silver in two directions, people could transport their money on their own with a cheque. Bankers issued and

accepted cheques from people in both cities, and the difference in traded value between Amsterdam and Brussels—the trade deficit—could be settled by transporting silver. The cost of money transports depended on the distance and risk (Cantillon, *An Essay in Economic Theory* , 1755).

These cheques or bills of exchange made payments between people more efficient. The vital fact to note is that those money transmitters came into existence because transporting the physical commodity was costly. Bankers had a job moving money from one place to the next because money was a material thing, and the risk was a physical threat. In the present, transferring money is a near-automatic update of a digital bank ledger. The costs are not about the physical risk of getting robbed or the time spent on the road with a bag of cash. Instead, modern bank transfers are costly because they have to win and maintain the trust of their customers. We have to trust the bank to maintain the ledger. They, besides the buyer and seller, are a *third party* to the transaction. Both the necessity and drawbacks of having a third party in the transaction are evident if we look at the history of money. However, cryptocurrencies address the drawbacks by making money-transferring third parties obsolete.

Money as Trusted Numbers

Imagine if all the people of a city could teleport money to their suppliers with a magical hand wave. In such a magical world, the job of transmitting money does not exist. The most magical money we can imagine is money that moves between owners with zero friction. Does modern electronic money create that magical world? It almost does. A payment done by PayPal, Transferwise, or SWIFT feels close to the magical teleportation of value. However, we still require a business relationship with a bank. They hold your deposit of digital currency. You have to trust them to execute the transfer that starts with your magical payment. If they fail to do so or have questions about your payment, your business is on hold. Ask travelers or people that run an online international company how these money transmitters make them feel. Many of them are unhappy customers.

Besides, what is the thing that banks transfer? This type of money is created by *fiat* , which means ‘decree,’ an arbitrary order, authorization, or sanction. Fiat money is created by the single sovereign authority of a nation-state because it *can* . It has value because they say so. You have to pay taxes with that money, and often people are legally obligated to accept

it as payment. Authorities enforce the scarcity of fiat money with laws, and they arrest counterfeiters. In contrast to commodity money, understanding the origin of fiat currency takes some study. Pre-industrial merchants had a visceral understanding of the gold and silver they accepted as payment. Precious metals were a commodity with a long history. People agreed on its uses. They could trust their available options to trade it for other things. The same day, the day after, or in a few decades, in retirement. In contrast, the numbers in a modern merchant's bank account do not behave like precious metals. These numbers lose a bit of purchasing power every year through inflation. The price of a currency relative to other currencies is unpredictable, in addition to the already variable price differences of goods between countries. Modern merchants and consumers understand *their* fiat money *too* ; most of them avoid keeping large amounts in their accounts. They invest a portion of their fiat money into other assets that will hold their value, and the remainder is for spending.

Whether we consider modern merchants using fiat for payments, or pre-industrial merchants using precious metals, we can be sure that they all managed their money according to their rational self-interest. And since fiat currency is very much unlike precious metals, its owners naturally behave differently.

In pre-industrial economies, bankers, merchants, and politicians had financial instruments other than metal money. As mentioned above, they did use cheques to settle trade, and letters of credit, shares, and banknotes. They used these instruments instead of money when necessary or convenient. However, they never confused these financial instruments with money itself, which was strictly metal. We can assume that most people understood why precious metals had value. They had an accurate and intuitive sense of the exchange rates between gold, silver, and copper. Their monetary landscape was much more straightforward than ours. Pre-industrial money was always a kind of physical stuff that behaved strictly according to the limits of physics, and they never confused it with the bills of exchange that represented the commodity.

Physicality and Origin of Money

We know that a range of commodities served as money throughout history, including seashells, gemstones, beads, gold and silver coins (Nick Szabo, *Shelling Out: The Origins of Money*, 2002). Nowadays, we have fiat money

in paper and electronic form. We can compare these types of money in at least two ways: the place it originates and its physicality. The commodities that served as money in the past were a natural part of the environment. For example, gems and metals come out of the ground, and people have to collect seashells from the sea. Acquiring these commodities took work. However, the ability to do this work was distributed across society and location. In contrast, modern fiat money belongs to the state and banks. The central bank creates the money to pay for the assets it buys, which are usually government bonds, which are debt that the government owes. Then commercial banks can lend out this created money again. It takes no work to create this money. The only requirements are:

- **Sovereignty** , that is, the ability to enforce the use of the fiat currency within a geographic region.
- The strict **regulation** of the banks, who keep track of people's balances.

In contrast to fiat currency, commodity money is created by the hard work of multiple entrepreneurs. The power to create fiat is central, while the ability to create a commodity is distributed. The scarcity of commodities is natural and robust. The scarcity of fiat currency is artificial and fragile. Since the invention of banking, money became less physical; it changed from a valued commodity into a record of a transaction. Most of the money supply is now merely records of who owes what to whom. By finding better ways of transmitting, saving, and investing money, bankers created kinds of money that are easier to handle. In essence, you can send money with letters of credit and bills of exchange, and you can create more of it by fractional reserve banking, in which only a small portion of bank deposits are backed by actual cash and available for withdrawal (Investopedia).

The characteristics of physicality and origin of money enable us to differentiate between the following kinds of money:

- physical commodity money mined and collected by entrepreneurs;
- paper money issued by a private bank or a government, backed by commodities it holds in deposit;

- paper fiat money issued by a central bank of a sovereign government;
- paper fiat money issued by local or regional cooperative;
- digital fiat money issued by the central bank of a sovereign government, which regulated banks lend out to companies and individuals;
- digital money issued by a private, entrepreneurial entity or group. Although examples of that kind of money have been tried and have failed, such as E-gold, private digital money did succeed with the invention of cryptocurrency, with Bitcoin as its first instance.

We can appreciate the power of cryptocurrency by looking at two recent examples of money that came close to the idea of digital, distributed money, but failed to achieve global distribution. Local fiat currencies are an example of distributed money, and e-gold is private digital money. Both failed the test of resistance to censorship, which reveals the unique qualities of cryptocurrency. Local fiat currency is printed by local authorities to stimulate the economy of a town or a wider region. Local governments meet with small businesses to print that local fiat currency. Examples include the Brixton Pound, BerkShares, Salt Spring Dollars, Bangla-Pesa, Ora, Kalentalese dinar, caribe, Fausto, etc. People can agree to accept this money as payment at the same value as the national currency. The small scale and trust allow them to act as a central bank to print money and boost local commerce. The national government sees these small-scale projects as mostly harmless and allows it. However, issuing private money as a company, such as in the case of e-gold, tends to be forbidden by national governments. E-gold was private, digital, commodity-backed money. That type of money scratches at the sovereign national currency. It also enabled money laundering and illegal transactions, which was the perfect excuse to shut e-gold down.

Fiat currency requires the highest political authority, a sovereign power, to enforce the acceptance and scarcity of the national currency. In the 20th and 21st centuries, before cryptocurrency existed, you had to trust the bank. You did not have a choice. That is why bankers complied with strict regulations, their offices feature granite, and they wear suits. Fiat currency relies on the political power of the sovereign and the fact that the public trusts banks. The sovereign state and banks are both necessary and

sufficient for fiat currency to work. Banks are comfortable in their control of financial markets and transactions, protected by regulations, which form a barrier to entry. Real capitalism is based on healthy competition, but with a monopoly on the field, the free market stops working. Business becomes political. The documentary *Saving Capitalism* by Robert Reich describes part of the situation in the United States. In Europe, similar voices are pointing at their own broken systems. Japan has been having monetary problems for decades. Since the Great Financial Crisis of 2008, the public has been feeling that something was wrong, but it takes time for researchers, artists, and producers to make those failings accessible to our layman understanding.

Bitcoin transcends the limitations of physicality, government control, and the origins of money. It has the attributes of a scarce commodity that requires work to be made, and it is digital like a bank ledger. And yet, it requires neither a sovereign who declares it money nor trusted institutions to transfer it. This new type of money enables economic transactions that we can not yet even imagine. What we do know is that from now on, our transactions no longer require that third party.

Money as Scarce Digital Digits

Local fiat currency, national fiat currency, and private digital commodity currency have failed in ways that show the unique strengths of cryptocurrencies: scarce digital tokens, issued and distributed by multiple entrepreneurial entities, through an elegant system of checks and balances.

People can use cryptocurrencies on a global scale. It is a commodity, transacted across the globe, by private companies and individuals. The features of cryptocurrencies are a digital version of what puts physical commodities in high demand. Since the launch of the Bitcoin network, your trust in bankers is no longer a required ingredient of a transaction. People can choose. To the extent that Bitcoin and other cryptocurrencies become more liquid, we have an alternative to banks. You can trust the banker and the whole regulated system, or you can trust an immutable blockchain. You can choose to accept or pay with fiat currency or cryptocurrency. Cryptocurrency is a candidate for making those payments and financial instruments the banking system fails at. We will cover these cases in later chapters. For now, consider that a bank has little reason to innovate. Their

methods are an easier, safer way to profit, without taking on risk or hard work.

The properties of money show that it requires trust in the stability of its value, for a given time horizon. Fiat currencies tend to be stable in the short and medium term, whereas gold and silver are stable in the long run. Whatever people use as money, it has to be good for something and scarce. Something can only be rare in a meaningful way if it also has value. A commodity can assume the role of money only after it is wanted for another reason first. In the case of Bitcoin, it had value as a digital collectible and an economic badge of honor. Owning Bitcoin implied that you could imagine the possibility of money that was better than both fiat currency and commodity currency.

Unstoppable Money

Private money can only succeed if the state cannot stop its issuance and transactions. It needs a quality that makes it unstoppable. We already know that governments want to prevent private money transactions: it is a challenge to the business model of public funding. There is a reason the state created fiat money: it gives them more power and financial control over the population. Historically, the need for that control was to defend the nation in war (James Dale Davidson & Lord William Rees-Mogg, *The Sovereign Individual*, 1999). In recent times, many nation-states have become bloated hierarchies that serve a class of bureaucrats and their dependents. Private money that circulates widely would challenge the interests of nation-states, so it can only succeed if they cannot stop it from being used as payment.

Bitcoin is unique in that it uses cryptography to create a digitally scarce asset, which, if you choose to, only you control with a private key on an immutable ledger. Bitcoin's immutability is a new quality. It has the advantage of no longer having to trust an identifiable authority, such as a bank. Bitcoin is unstoppable and immutable because it is decentralized. Its immutability depends on the proof-of-work algorithm for distributed consensus. Bitcoin is a network that runs on many physical computers ("machines") in many different physical locations. All are working with the same ledger with blocks of the same transactions. They all follow the same rules for achieving consensus, which automates the issuance and scarcity of Bitcoin. Because Bitcoin runs in so many places, there is no single 'place'

anywhere where anyone could go to stop it. As long as the system works as designed, it is unstoppable, immutable, and uncensorable. That was a radical disruption. The immutability depends on decentralization, which in turn depends on a consensus algorithm, which, in the case of Bitcoin, depends on proof-of-work.

We know that private money has to be decentralized because there is a long history of people issuing private currencies. At first, it used to be done by private banks, in the era of free banking, until the introduction of government fiat currency. In the present, some towns issue a local currency. The first digital currency that made use of cryptography was DigiCash, and the first private digital currency in circulation was e-gold. As mentioned above, the U.S. government shut down e-gold. This tendency to forbid and stop private digital currency was the reason why in 2009, Satoshi Nakamoto, the pseudonymous inventor of Bitcoin, focused on the way to decentralize private currency. The new digital currency had to have multiple nodes that could coordinate without needing to trust any individual node. That is why he had to solve the problem of reaching consensus. He could only invent Bitcoin because he found a practical solution to ‘The Byzantine Generals Problem,’ which deals with the problem of ensuring a trusted consensus with participants that are not to be trusted. That was the audacious goal he took on and reached. The solution enables tokens that are digital *and* scarce at the same time. His Bitcoin solution and software did not get much attention at first because almost no one cared about non-fiat currencies at the time. Even most of his peers in cryptography and programming did not pay much attention. To paraphrase Schopenhauer: Nakamoto hit a money target that few could even see. Solving the Byzantine Generals Problem was a multidisciplinary achievement. That is why it is so easy to go off-track when making sense of Bitcoin. A casual investigation of Bitcoin will overlook the reasons for its functioning in this way.

The Politics of Money

With a historical sense of money transmission, and knowing that Bitcoin is unstoppable money, we can appreciate the political factor of money. This new angle enables a fresh entrepreneurial opportunity.

The key qualities of Bitcoin are:

1. Scarcity: it is hard to create;
2. User-friendliness: it is easy to hold, send, and receive;
3. Supply predictability: no one can inflate beyond the issuance schedule;
4. Freedom: you can use it without permission;
5. Resistance to confiscation and censorship: it is difficult to seize.

Cryptocurrencies had to be private money because it is unlikely that governments benefit from issuing a decentralized cryptocurrency. They can issue digital fiat currency already. There is no point in using a blockchain with a consensus algorithm because digital fiat currency does already function according to its purpose. Governments can only benefit from a decentralized currency indirectly, at a later stage of adoption, in case the top cryptocurrencies become viable assets for central bank balance sheets. The first central bank that expands their portfolio into cryptocurrency will have a first-mover advantage. If other central banks also buy, the price will rise further to a new and more stable level, securing a profit for the first mover.

Cryptocurrencies and crypto-assets are the foundation of a new financial system. An entrepreneur can build a company on top of a money system without having to talk to bankers. Note that we are still at an early stage. These new types of financial services can only grow with the liquidity of the cryptocurrency. Your customers have to see crypto-assets as a regular part of their life so that using them causes zero friction. Most people do not use products and services that frustrate them. No technology reaches the mainstream until it is guaranteed to be easy to use for almost anyone. In a world where cryptocurrency is widespread, your company can offer a customer faster and easier service. This new financial system *competes* with fiat currency. Governments can only benefit if they cannibalize their own fiat system. At the time of this writing, there are a dozen central banks who have plans to launch a 'blockchain version' of the national currency. In the opinion of many crypto enthusiasts that I have met, this scheme can not work. Fiat currency is already digital; there is no need for a blockchain, any decentralized ledger, or consensus algorithm. These techniques only help currencies that *lack* the authority of a nation with a central bank. The proposition of Bitcoin described above is against the status quo; the political and economic elite has failed the public with regard to managing

the money supply. According to crypto enthusiasts, they have betrayed our trust.

Trusting the government is like trusting a parent—they may or may not deserve it. Just as psychologists have different models and theories about parenting, politicians have different ideas about organizing society and the powers of authority. A good-enough government and legal system will have the power of authority, but only with the “consent of the governed,” according to the Universal Declaration of Human Rights.

Bitcoin, the first cryptocurrency, is a charged political project because the creators and most of the innovators of Bitcoin reject the authority of any government over the creation of money. Although different types of activists vary in their political views, those behind the growth of Bitcoin maintain that governments have a terrible track record in handling the supply of money. All political authority has failed or will fail the test for being a ‘good-enough government’ within the realm of monetary policy. Financial history is riddled with instances of capital controls, asset confiscation, currency debasement, hyperinflation, and sovereign default (J.T. Reed, *How to Protect Your Life Savings from Hyperinflation & Depression*, 2010). That is why anarchists and most libertarians believe that most or all parts of the government do not deserve your trust and support. They claim that anything the government does for you comes at your expense, or at the greater expense of someone else. In contrast, liberals, democrats, republicans, and conservatives believe that governments deserve their place in the present and the future and that their officials will exercise sovereignty well enough. That includes the money system as it is. Politicians like fiat money. More specifically, they like the ability to create more money at little cost; they like taking on public debt, and the central planning of the interest rate. The official narrative around fiat currency presents the monetary status quo as necessary and proper. Mainstream political action depends in part on fiat money, but the official story hides this critical fact. Modern sovereignty is a condition for the ability to print money.

While fiat currency requires the coercive power of a state, decentralized cryptocurrencies can resist that coercion while competing as a monetary good. This difference of political opinion between the mainstream and the creators of Bitcoin will shape the path of adoption for cryptocurrency

because it unleashes open competition for monetary goods. Liquidity for crypto-assets weakens the monetary pillar of the government. Before Bitcoin was a monetary good, anarchists and libertarians only had debate and rhetoric as tools of persuasion. After Bitcoin, no more debate is necessary; they can act on their conviction by holding crypto-assets. They can exit the financial status quo. A pointless discussion between a minority and the mainstream has become a material contest, to be fought with disruptive money. Cryptocurrency opens the door to exit the fiat monetary system. Whoever takes that exit, stops supporting the political mainstream. With every holder of cryptocurrency, the anarchist and libertarian case gains strength. As the decentralized power of crypto-assets grows, fiat currency weakens. If that process continues, then at some point, the weaker authority will start to fail at regulation of the companies that build on top of cryptocurrencies. This could start a feedback loop of failing government and rising influence for the holders of crypto-assets.

Governments, as sovereign entities, are not usually thought of as a competitor. For example, the monopoly on violence is only challenged by other nation-states by waging war, or by risk-takers with debatable ethics, also known as criminals. Likewise, Bitcoin challenges the government's monopoly in the domain of money and finance. Any financial operator has the option to compete with the government and banks. Before Bitcoin, printing your own money was not done. Even if you did, you had to keep the scope limited, like with local currency or currency for use inside a multiplayer game. Ambitious attempts at money, such as e-gold, were futile because they could easily be stopped. However, Bitcoin could—and most likely can—not be stopped, so it is the first currency to have a chance to challenge the sovereignty of the state. It was naughty and edgy. It was a new way in which political minorities could exit the financial status quo.

Up until 2019, Bitcoin was not subject to overt attempts to stop it. One reason is that most government officials and central bankers have not made up their minds about it. Even if they are vaguely against it, they do not know what they are looking at. This is obvious when you see a parliamentary committee investigate cryptocurrency by interviewing central bankers, cryptocurrency practitioners, and traditional finance types. Since late 2019 and in 2020, several U.S. politicians say they want cryptocurrencies outlawed because they undermine the interests of the U.S.A. They know what they are looking at: much of the geopolitical power

of the U.S. comes from the U.S. dollar as an international reserve currency. However, most politicians and central bankers do not understand the vision for cryptocurrency. They take their old yardstick for deciding whether it is money and see that cryptocurrency does not resemble fiat currency. They call it a digital commodity, or an entirely new asset class. Within the frame of their job, they are correct. It is indeed a new asset class because there is nothing else like it. It is also a commodity, undoubtedly similar to gold, and gold is serving as a mere commodity or as central bank assets. But in the past, gold was money. It started as a commodity, became money, and then went back to being a commodity for most people and only money-like for gold holders and central banks. Likewise, a scarce digital commodity can become money. It has all the necessary attributes to become money, plus a few extra ones, such as the ability to send it from one smartphone to another. Elite bankers and politicians tend to view the surface attributes of cryptocurrency correctly, but their frame is too narrow to see the big picture. Another possible reason that most governments do not try to stop Bitcoin could be to prevent a path of rebellion from crystallizing against their policies.

A rebellion would empower the ecosystem and make it even more robust. It seems that most of the elite bankers and politicians fail to grasp cryptocurrency. It is nearly impossible to keep up with the developments in Bitcoin, other blockchains, and crypto-assets. Understanding how it works requires basic knowledge of a dozen branches of science. From there, you can dive into the details, which all come from specialized technical fields. A traditional politician, finance MBA, and even many conventional technology investors are not in a position to gauge cryptocurrencies accurately. They are too busy rendering crypto-assets through the lens of their own job, background, and responsibilities.

What Cryptocurrency can Become

According to some people, Bitcoin, and perhaps other blockchains, can secure all the value exchanges in the world. Any financial instrument that moves value around could become an automated contract on top of a decentralized digital currency. You can have multiple platforms on top of a cryptocurrency that keeps various types of arrangements secure, with different privacy and security features. Any 'contract' about a 'financial product' can then control a specific sum or flow of money. Those who want

to deal with that kind of payment will be early adopters. As the demand for that money grows, more contracts and products become available. It becomes easy to water down the security for individual contracts and assets by having them run on an independent but smaller blockchain. The market will converge on the level of security and settlement assurances for each type of asset. Each product or service on top of a cryptocurrency will have to reach a balance between constraints. One way of defining these constraints is *security* , *convenience* , and *freedom* (Dan Geer, *Cybersecurity and National Policy* , 2010). These constraints form a triangle from which the designer picks two of the three for a given system. Each of these balanced security solutions has to be built by a company. Maybe that company could be founded by you.

It seems that Bitcoin and other cryptocurrencies can already hold their own in parts of the economy. We can expect future functionality to absorb a broader range of transactions. We will depend less on institutions and will have less exposure to the arbitrary central planning of the money supply. Entrepreneurs will be free to create exactly the financial services that the global population demands, without having to jump through hoops set by regulators. Building on Bitcoin prevents dealings with people who must comply with regulations designed for a world in which the transport of value requires a third party. That freedom to disrupt global finance will meet resistance from the existing institutions because money is a piece of nation-state politics.

CHAPTER 2

The Political Entrepreneur

WHATEVER YOUR AIMS ARE as an aspiring crypto entrepreneur, your success will disrupt the status quo of money and finance. They have power: these organizations are in control of their situation and can influence their future to their benefit. You will succeed in your mission and maintain the health of your organization if and when you replace that status quo. Your success is predatory; it lives off energy taken from the incumbent. We have to study the relationship between entrepreneurship and political power so that we are aware of the type of pressure that regulators and banks could apply to a crypto-organization.

Notice how unusual it is to use the word “power” in an entrepreneurial context. Power is a political concept, and traditional startup companies have little to do with politics. Old-school entrepreneurs take the laws and regulations as a given and play it as it lays. Most entrepreneurs stay clear of the friction that comes from going against the political grain. Only the most successful entrepreneurs whose technology influences societal trends, activism, and election outcomes become politically relevant. If the politicians already have issues with a disrupted media landscape, as we saw in the case of Facebook and Cambridge Analytica, we can expect that conflict to escalate with the introduction of disruptive money. Therefore, a crypto startup is more likely to succeed when it engages political power as part of its business model. You are going to get customers who used to be captives of elite institutions. The new crypto-assets that a startup creates and earns for its stakeholders take power and wealth away from traditional financial institutions.

Free Market Technology with Political Teeth

The internet is a catalyst for the general trend of decentralization that started after WWII. At the time, most of society was organized in large corporations and large governments. People enjoyed a uniform culture broadcast by mass media such as TV, newspapers, and radio (Tim Wu, *The Master Switch* , 2010). Information technology eroded traditional media

such as newspapers and TV, while empowering marginalized groups and individuals through message boards and social media. We now live in a 'custom-made' society with differing lifestyles, cocoons, and echo chambers. Western societies especially are highly democratized, with power distributed among a large number of institutions and constituencies. Information technology tends to make society more open and flat, less hierarchical, and it tends to develop in cycles. First, an inventor makes a new medium available. As entrepreneurs learn the commercial angles of the medium, companies fight over the market, until the information flows from producers to consumers through corporate channels, curated and censored by the political status quo.

Software is eating the world (Marc Andreessen in the Wall Street Journal, *Why Software is Eating the World*, 2011). For example computers connected by the internet have eaten snail mail, newspapers, radio, TV, music, and books. Money and the financial industry were immune to that trend for a long time. Since the Bretton Woods Agreement of 1944, all the money in circulation depended on central political power. The Bretton Woods System was backed by gold but ended in the early 70s when the U.S. stopped exchanging gold for dollars. However, the political foundation of currency remained: since then, in almost every country, the central bank creates fiat money which is used for payments as physical paper or electronic transfers between licensed banks. People trust fiat money for payments because of that political backing. Creating money was a privilege of sovereign entities: to win trust, the organization needs the political power to claim and defend the sole right to print money. Entrepreneurs had no reason to look at money and finance as value propositions because that political dependence ruled out competition from private entities. Law enforcement authorities have and will shut down any company that creates a private digital currency that succeeds. There was no sense in undermining the sovereignty of a state, so the software industry did not attempt to build products that "eat" money and finance.

Shutting down a successful cryptocurrency is either impossible or prohibitively expensive. Bitcoin has a track record of immunity to political meddling; it is slippery on the defense and an indirect threat to any central bank or government that mismanages its money supply and public finances. Furthermore, banks are no longer the gatekeepers of financial progress. Any startup can offer a financial or payment service on top of a cryptocurrency.

As the liquidity of cryptocurrencies grows, these new financial services will draw customers away from banks and perhaps even insurance companies, investment funds, and other financial institutions.

A Cryptocurrency is a Virtual Jurisdiction

Three technical problems prevented the issuance of private, decentralized currency: the double-spend problem, decentralized consensus, and provably fair issuance of new money. Before the invention of cryptocurrencies, these obstacles prevented a guarantee of scarcity, which is needed for people to trust an asset as money. The launch of Bitcoin opened a new playing field. Private parties can now rely on *software* for transacting and storing value. Digital scarcity gives anyone a choice: you have the option to transact with fiat currency on closed corporate networks, or decentralized cryptocurrencies and exchanges. That choice will erode the power of banks.

A crypto entrepreneur has to understand the role of political power in the money system. As it is, let us assume that political action is not a standard piece of your strategy. To a typical entrepreneur, political action looks like an unrewarding game. The type of person who starts a company is unlikely to consider a career as a politician, with the exception of established billionaires. On the contrary, an up-and-coming entrepreneur believes in free markets, voluntary transactions, plus-sum games, open innovation, satisfied customers, and profits earned by fair competition. These values have little to do with political action. However, the power of fiat money is political, to disrupt it your information technology needs the teeth to match and exceed that power.

You could say that an immutable blockchain is a global, voluntary, virtual jurisdiction. You could also call it a 'smart jurisdiction' for 'smart contracts.' Bitcoin and Ethereum are not in a specific place; they function on the internet, as a global and distributed network. On a decentralized system, people transact, but their assets do not reside in a particular place. The people, when identified, might be subject to a traditional jurisdiction or another, but this is new technology, and the way the rules apply is vague, new, and most likely not defined. That is an advantage for people who choose to take responsibility for all the consequences of their transactions, both good and bad. Whatever financial game you are playing, there is no automatic authority there. The field is free and open. With the narrow exception of earned income, if a regulator has an issue with your

transactions, the burden of discovery and investigation lies with them. You do not have to ask for permission; instead, they have to come to you and explain the rules, should they even bother. For example, if international travelers memorize the keys of an undeclared wallet with one million dollars worth of cryptocurrency, are they breaking the rule that says you must declare cash worth more than ten thousand dollars?

Sovereign entities at odds with the established banking system already use crypto-assets—to apply an idea by Von Clausewitz—as a continuation of politics by other means. If politicians in North Korea and Iran achieve their ends using crypto-assets, should not others be able to do it too? As ordinary people, we are not used to thinking and acting as sovereigns, but we can be open to the possibility of free currency.

Building on Decentralized Money

As Bitcoin and Ethereum acquire more ‘law’-like capabilities and more functionality for fundraising, they can compete with nation-states that rely on traditional laws. They become virtual jurisdictions: nation-like entities that compete with nation-states. Entrepreneurs, as agents of these sovereign entities, have to practice diplomacy. This territory of freedom has an analogy in the separation of church and state, which happened around five centuries ago in some places (Erik Voorhees in Bitcoin Magazine, *Separation of Money and State is the New Separation of Church and State* , 2015) The Catholic church lost exclusive authority over its territory of religion, and also lost its jurisdiction over physical area to nation-states. Likewise, the nation-states and banks will lose their influence over money and finance (Martin van Creveld, *The Rise and Fall of the State* , 1999).

We already have historical examples of decentralized money: gold, silver, and banknotes issued by private banks. That decentralized money disappeared when governments and banks founded central banks to issue money. Cryptocurrency will reintroduce decentralized money, with the added benefit of near-instant global payments.

Fiat currency works similarly to banknotes from private banks, but it is backed by government debt. A central bank buys financial products on the open market, usually government bonds. It pays for these bonds with money that it creates. Some of the money is paper cash, but most of it is numbers on a balance sheet. That means that the government bonds become assets on the central bank’s balance sheet, and the created money is a

liability on their balance sheet. The central bank lends this money to private banks at interest. When it sells the bonds, and it receives fiat money back, it can destroy that money. That is how it controls the total supply in circulation. Other private regulated banks can also issue money through ‘fractional reserve’. They can lend out more money than they receive from the central bank or customers because they can expect most of their customers to keep their money in the bank. See Investopedia for more detail on central banking and money creation.

The system of fiat currency relies on the trust of the public. The central bank can only circulate money that people trust will hold its value well enough. And ordinary banks make loans while keeping a fraction in reserve. Most of their customers keep their money in the bank, and only a small portion of these customers withdraw their money at any given time. A bank can maintain that fractional reserve as long as their customers trust that they can withdraw their money at any time. When that trust breaks down, you get a so-called *bank run*. In case a large portion of the customers attempts to withdraw money, and the bank does not have enough fiat cash on hand to fulfill those obligations, then that bank needs the central bank or the government to act as a lender of last resort. If no more significant lender is available to cover the ‘run’ bank’s obligations, then it goes bankrupt. It is a sophisticated but centralized system. Crypto enthusiasts work on a future where banks are no longer necessary. Cryptocurrencies and smart contracts will replace banks, especially central banks who create fiat currency. We can expect cryptocurrencies and crypto-assets to inherit the features of the banking sector, and then we can see a boost of global finance by orders of magnitude. These new companies will share the benefits of this new financial system with their customers, which will eventually include everyone. Cryptocurrencies empower a slow but deliberate run on all the banks.

The Territory of Freedom

As an entrepreneur who nibbles at the power that comes from sovereignty, you have to remember: It is not what you do; it is who you piss off. Satoshi Nakamoto said that Bitcoin gives us, the users of cryptocurrency, a “few years” of freedom. He acknowledged that you would not find a solution to political problems in cryptography, but “we can win a major battle in the arms race and gain a new territory of freedom for several years.” (Phil

Champagne, *The Book of Satoshi: The Collected Writings of Bitcoin Creator Satoshi Nakamoto* , 2014) This “territory of freedom” is an essential idea for crypto entrepreneurs. Which technologies expand this territory of freedom? Technologies that support privacy, the interchangeability of assets, and the removal of friction in communications or transactions. As Satoshi said: “Governments are good at cutting off the heads of centrally controlled networks like Napster, but pure P2P networks like Gnutella and Tor seem to be holding their own.”

Bitcoin’s first freedom test was the operation of Silk Road, “the eBay of drugs,” where decentralized money proved resistant to coercion from regulation and law enforcement. People traded illegal goods on hidden websites, and they still do. The founder of the Silk Road, Ross Ulbright, did get caught because he bit off more than he could chew. He pissed off law enforcement agencies so much that they unleashed more force against him than his operational security could handle. Other early Bitcoiners, such as Charlie Shrem, battled with financial regulators. Exchanges that used to host anonymous trades are now under pressure to make their customers known. Not all companies that rely on a blockchain can defend the freedom it affords. However, the Bitcoin network itself has survived its trial by fire. So far, it seems that no one can stop this entity. Understanding *why* Bitcoin is uncensorable is an important part of your education on crypto-assets.

Any entrepreneur who offers a product or service in a gray area has to prepare for battles with regulators. Exchanges, crypto-asset ATM providers, over-the-counter traders, and issuers of tokenized assets all operate in this gray area. Are you going to operate under a known brand, above the radar? Then your venture needs specialized legal advice before you even get started. Your executive team needs to include an attorney with skin in the game. An alternative is to keep your activities so far below the radar that regulators will not bother stopping you. Then you take advantage of the innovation made possible by cryptocurrency without raising the risk. For example, small but professional over-the-counter traders can make a living holding an inventory of crypto-assets. According to the law in most jurisdictions, they need a money transmitter license and comply with KYC/AML regulations. However, because the amounts they trade are either small or below the radar, they are not worth the trouble for regulators. They can continue working because, so far, they do not piss anyone off.

Another example is issuing a token without registering with the regulator. Some people run a project that is so small that they are willing to take the risk of unregulated business, rather than accepting the cost of getting permission. In case a startup company wants to sail close to the wind concerning rules, again, it is best with an attorney on the team.

In contrast to Bitcoin, Ethereum has not yet been battle-tested by a conflict with a nation-state adversary. The founders of Ethereum are not pseudonymous, and they organized themselves in companies and foundations. Critics of Ethereum doubt whether it can withstand an attack from a nation-state actor. Ethereum needs a Silk Road-moment to show that it can resist regulators and law enforcement agencies.

How Much Security Do You Need?

Blockchains that are smaller than Ethereum face additional risk. They are secured by less proof-of-work mining, or experimental mining algorithms so that ordinary hackers can attempt so-called double-spend attacks, which undermine the trust of the user in the victimized blockchain. Also, these projects have fewer developers inspecting the code for bugs and exploits. An immutable and secure blockchain is expensive on multiple levels: energy invested for proof-of-work mining, open-source leadership and management, and attracting new developers and entrepreneurs to the ecosystem. It seems that a blockchain needs to grow big and stay on top to be immutable and safe.

The creators of Bitcoin designed the network for a purpose that they expected would piss off nation-state actors; the security of proof-of-work mining matches that threat. By many standards in finance, Bitcoin is the most secure way of moving and storing money. The game theory of mining, buying, and selling keeps the cost of security accurate: if the price of Bitcoin rises, mining becomes more profitable, which attracts more miners. Each new mining rig invests more electric energy in the security of the chain. Although some people call this proof-of-work security inelegant and wasteful, we can view Bitcoin as a fair marketplace that transforms electrical energy into financial security.

Ethereum's level of security is still an open question because its use of proof-of-work mining is reluctant. The leadership considers proof-of-work a temporary solution while they develop a secure proof-of-stake model, which does not cost electricity or hardware but allows owners to earn fees

by making blocks of transactions. As mentioned in the previous section, another vulnerability is the public nature of the organization that develops Ethereum. The development team is organized in a foundation and seems more hierarchical than that of Bitcoin. The legible organization could face pressure from regulators or politicians. Even though the nodes that run the network stay independent, some actors could pressure the team into building features and backdoors they want. If Ethereum fails through censorship, that will validate the proposition of Ethereum Classic, which has a fixed issuance policy, a future dedication to proof-of-work, and no centralized foundation. Ethereum Classic would then attract those startups that need a ‘naughty’ smart contract, resistant to powerful would-be censors.

Although Ethereum might fail under the pressure of a nation-state attack, it could still have enough security to support a set of less significant financial applications. Non-controversial companies could still rely on Ethereum’s blockchain. Governments and regulators might not bother to apply pressure to prevent these smaller disruptions. A startup can account for the means and motives of its adversaries, and seek protection from those threats. Another category of crypto-assets, gaming assets, require even less resistance to censorship. Ordinary assets, which do not offend the powers that be, only need settlement assurances for the transacting parties. Ethereum and other blockchains can do that better than middlemen or institutions. There is no need to over-secure every blockchain asset against censorship, which suggests a future with multiple blockchains.

The security of assets depends on costs, risks, and rewards. We get hold of things we value and keep them secure. Whether these things are material or intangible, these costs and risks are physical. The challenge is to find the right security solution for each of our valuables. Consider a simple metal lock. People who often ride a bicycle or motorcycle have an intuitive grasp of the lock their two-wheeler needs. In most cities, a new or classic motorcycle of a high-end brand needs a lock worth at least \$100. A determined thief with a diamond-cutting tool can open that lock in ten minutes or less; however, if the prize is parked under a streetlight in an open space, then the thief will not get that opportunity. A bicycle with a price tag of \$200 is usually safe with a \$30 lock, but only when secured to a hard point such as a lamp post or fence. Some bikes are so rusty that they are not

worth stealing. In some jurisdictions, authorities punish vehicle theft harshly, so that you can park a motorbike without an additional lock.

Each piece of property has a threat model, and sooner or later, the owner becomes familiar with it. To keep your crypto-assets secure, you merely apply that logic to your digital environment as well. Digital assets have digital locks and keys to protect against thieves with digital tools. We will look at the specifics of digital wallet security in a later chapter. For now, we consider that securing information is, at the foundation, the same as securing physical objects.

The up-and-coming wave of crypto companies will contribute to the translation of physical security concepts into the digital sphere. Once that translation is complete, crypto-assets will have all the advantages of being digital, yet none of the drawbacks. At that point, they will be part of ordinary life as trusted utilities in the background that we use without much effort. The crypto-assets secured by cryptography, running on a blockchain-based virtual jurisdiction, emerge as a new normal that promotes a redistribution of power from central actors into broad communities of peers. The decentralizing and fragmenting forces that started on the internet are going into third gear with crypto-assets.

CHAPTER 3

Liquidity and Adoption

THE TOTALITY OF COMPANIES AND INDIVIDUALS dealing in crypto-assets make up an ecosystem that is growing in size and maturity. Markets die without liquidity, so we need to grasp how the liquidity of cryptocurrency works. The trading volume grows because more people exchange their fiat money for crypto-assets and trade one crypto-asset for another. And the number of company relationships grows because each new company or market can expand the size and scope of possible deals. Just like in the natural world, this ecosystem develops and moves from one phase to the next.

For example, traders create demand for exchanges, which creates demand for market makers, which together create demand for data and software specialists, which creates demand for analysts, training courses, and conferences. New crypto-assets can create new industries, such as the smart contract that defines a basic token on Ethereum, the ERC-20 standard, which gave rise to a boom in token sales, or initial coin offerings (ICOs), which in turn created the demand for marketing services. The invention of so-called stablecoins—which are tokens pegged to a fiat currency—has created a market for borrowing and lending with collateral and leveraged speculation. Furthermore, it is now possible to put digital items from online games on a blockchain, in the form of a unique token. Gamers and speculators trade these tokenized in-game assets on the open market. We can expect that the inventors of new types of assets will create other new markets. As an entrepreneur, you fit your mission in this maturing ecosystem. In chapter 6, we discuss tokenized holdings in more detail. For now, we appreciate these examples as evidence of progress; crypto-assets with active users prove that this industry is *not* just a speculative bubble or a science experiment.

Even though this technology is in its infancy, we have come a long way already. When I discovered Bitcoin, at the beginning of 2013, you needed technical expertise to own bitcoins. Also, you needed luck; you could only buy or sell bitcoin in some locations, and the only public exchange market for bitcoin was Mt. Gox, based in Japan. I wanted to buy some bitcoins as

soon as I heard about it, but I gave up when I foresaw the hassle of getting my money, Thai baht and euros, into a Japanese bank. Furthermore, I was too worried about the security of my wallet setup. I felt I had to learn more about wallets before I could trust myself to keep bitcoins in the long term. A bit later on, I did a writing job for someone who offered to pay in bitcoin, but I declined because I saw no way to spend it. These challenges for the liquidity of bitcoins are now over, at least for people like me. Handling crypto-assets gets more comfortable with time so that it becomes accessible to a broader range of users.

The overall growth of these markets for crypto-assets has three limiting factors, analogous to physical friction inside a mechanical system:

- **On-ramp and off-ramp** —Trading fiat currency for a crypto-asset is similar to a car entering a highway via an on-ramp. The traffic on the road is smooth and steady, and the on-ramps and off-ramps are where cars can match and adapt their speed to the other traffic. When anyone wants to trade fiat currency for a cryptocurrency, it slows down to the speed of fiat. When you use cash, you have to meet in person. In case the traders do not know each other, they have to build trust or consider risks to personal safety. When you transfer fiat money electronically, your trade must comply with your bank's rules for 'anti-money laundering' (AML) and 'know your customer' (KYC). They tend to force you to limit your transaction size and demand explanations of the source of the funds. Regulatory authorities require them to do this as a condition for keeping their license to transmit money.
- **Usability** —The experience of trading, sending, and receiving money and crypto-assets determines what kind of person can participate in the ecosystem. For any given case or type of person, the process of completing a trade or transmission has a certain number of steps, and each has a level of difficulty or ease. Is this person forced to think? Which concepts does the user interface invoke? How long does it take? How much training do they need? Can they show others? Each of these elements adds up to the total amount of friction in moving money and assets.

- **Liquidity** —Every asset has a given number of buyers and sellers at any price level, which determines how easy it is to buy or sell. For example, selling or buying a house takes weeks or months, because it takes work to find the people willing to take the other side of the deal. It also takes time to compare that deal to other options in the same price range. Cars are a bit more liquid, but the dynamics are similar. In contrast, when you trade one fiat currency for another, you have almost no friction because there are so many people selling or buying that there is no effort at all to find a buyer or seller. Money is the most liquid asset in the economy. Cryptocurrencies and other crypto-assets have fewer buyers and sellers than fiat currency so that the liquidity still resembles that of commodities like foodstuffs or metals. Crypto-assets that trade at a low price or low volume are less liquid, which adds to the system's friction.

You can view each source of friction as an entrepreneurial challenge. Whether you comply with or run circles around regulators, you boost the on-ramp and off-ramp. Software that automates steps of a trade, or invokes better metaphors, will make crypto-assets accessible to new users and open new markets. Whoever connects pockets of buyers and sellers “makes a market” and boosts liquidity for that asset.

We can compare this ecosystem to the financial markets of the 19th and 20th centuries. It seems that crypto-assets can be at least as sophisticated. It could follow a trajectory similar to the internet, which decentralized publishing and communication, while matching and then exceeding the sophistication of traditional methods. The growth-potential of crypto-assets seems to resemble that of the internet at the end of the 20th century. There is a lot of work to do and many rewards to reap. As a crypto-entrepreneur, you can expect the 2020s to be roaring. Whenever you play around with a crypto-asset and feel irritation, boredom, confusion, or frustration, you are looking at an opportunity to make a difference and make a crypto-fortune. Entrepreneurs seek to solve a hard problem and to deliver the solution to a fast-growing customer base. What are the critical issues in the field of crypto-assets? Topics that insiders often discuss are:

- **Usability** —how to make transactions easier.

- **Scaling** —how to boost the transaction capacity of a given blockchain network.
- **Private transactions** —what will people do with untraceable money.
- **Interoperability** —how to exchange data between a specific blockchain and other software protocols and applications.
- **Regulation** —who decides who is allowed to do what with which crypto-assets.
- **New applications** —how to apply the insights behind Bitcoin to non-financial fields, such as corporate governance, politics, and nonprofits.

Most of these critical issues cause friction for users of crypto-assets. For a startup, they are challenges that require a solution. It is up to the team to build that solution as a product or service. Any company that executes well with a fitting offer can claim market share from banks. Every progress in crypto-assets will lower friction and advance the market toward mainstream adoption. Better on-ramps, off-ramps, and user experiences attract new pockets of users, which boosts liquidity, making crypto-assets more attractive. The industry as a whole is a self-sustaining positive feedback loop.

In the medium term, it seems that demand is lacking in groups of potential users, such as investment funds, online shoppers, and retail shoppers. The remaining chapters of this book will give you an angle of attack on these critical problems. Solve one or more of them for specific users of money or a financial product, and you have a good chance of turning your startup into a thriving company that competes with the traditional banks.

Connecting Crypto-Circles

Many Bitcoiners expect Bitcoin to progress through three stages of adoption: First, it behaves like a commodity that becomes a store of value, like gold, bonds, and real estate. People buy it to hold on to it to protect their wealth. As bitcoins become more accepted, it reaches the second stage: it becomes a medium of exchange, that is, a payment method for goods and services. If the liquidity ever equals that of fiat currencies and

the value is stable, it can be the unit of account, meaning that people will calculate prices of goods and services in bitcoin.

Some Bitcoiners already pay bills in “sats,” which is an abbreviation of ‘satoshis,’ the name of the smallest unit of bitcoin that you can send. You can divide a bitcoin into a hundred million satoshis, which seems like a ridiculously high number. The developers chose this amount for a possible future in which Bitcoin reaches the last stage of adoption and becomes a unit of account. In that world, the price of an individual bitcoin is very high. When an enthusiast pays a restaurant or hotel with sats, they prepare for that future of broad adoption of Bitcoin.

In summary, a commodity becomes a currency when you can save it, pay with it, and can set a price in it. In the present, Bitcoin and other top cryptocurrencies work as payment methods only among those that want it as a store of value. They see Bitcoin as a better way of saving and investing, and they welcome the chance to get more bitcoin. They save it for keeping their wealth liquid but secure, with the add-on benefit of high upside, because many believe that on the medium- to long term, the price of bitcoin will rise with the growth of the ecosystem. Companies will spend bitcoin and other cryptocurrencies if suppliers accept it, and their customers can pay them with bitcoin. Likewise, as a consumer, you are more likely to want to earn it when you know where to spend it.

These activities of earning, saving, investing, and spending can become a full circle. A commodity can only grow into the functions of a currency when it completes that circle; when there is no doubt in anyone’s mind that they can make good use of the currency because someone else will accept it as payment. These complete circles of transactions are, for now, small and isolated. When these circles grow and connect, the price of bitcoin rises, which boosts liquidity and demand. That is why companies that exchange crypto-assets have been the most profitable so far. The analogy to the natural world is *pioneer vegetation* : A growing ecosystem starts on barren soil without much nutrition. Only a select few species of plants can thrive. The plant matter from these species is the foundation on which other species can grow. Likewise, crypto exchange companies, traders, and market makers are benefitting from the early stages of crypto-asset development because fiat currency still rules. They are the foundation on which other crypto species grow.

The pockets of buyers, holders, and sellers start isolated, and exchanges connect those circles. But their current limitation is that they only make markets for the fiat and crypto-asset pairs. Most traders, speculators, or spenders start with exchanging fiat currency for crypto-assets. Exceptions are proof-of-work-miners, who exchange electricity for a cryptocurrency, and those who accept cryptocurrency as payment for goods and services. The crypto economy is a fiat-to-crypto exchange. The companies that help people sell their fiat for crypto-assets are the most successful, so connecting the circles of crypto-only trading is the core business of most asset exchanges. They win customers when they are good at trading across the boundary between fiat and crypto-assets. Examples of those companies are:

- **Crypto-to-fiat debit cards** —you get a debit card like any other, except that the asset you spend is a cryptocurrency. In practice, that means that when you pay for something with this card, the company who issued the card sells a portion of your crypto-asset in exchange for the fiat currency of the bill that you just paid. They take a percentage over that transaction.
- **Specialized trades** —There are specific markets that will pay a premium for a cryptocurrency, or willing to sell something for it at a discount. For several years, Pulse.io offered a way to get a substantial discount on Amazon products if you paid with bitcoin. Some people earn Amazon gift certificates as pay or bonuses. They prefer a liquid asset to the items they can buy on Amazon, and they need to trade it with those who want to purchase items at Amazon. Bitcoin is the perfect currency for that exchange. Pulse.io has now stopped operation, but it is a great example.
- **New payment processors** —Entrepreneurs sometimes discover demand for making a specific type of payment. A current model is Bitcoin wallet company Zap's application which lets customers pay a webshop with fiat currency, while the merchant receives bitcoin. This payment bypasses the friction of trading fiat for cryptocurrency and vice versa. Insiders have often observed that making cryptocurrency invisible is a significant step toward driving adoption, and Zap is an excellent example of that.

- **Stablecoins** —Ethereum and other smart contract blockchains enable the transfer of tokenized assets, and one of those assets can be a fiat currency. Another good name is “crypto-dollars.” We can expect other tokenized fiat currencies later on. These crypto-assets have a stable value like fiat currency, and you can transmit them as cryptocurrency. In case the holder wants to limit their exposure to the volatility of crypto-assets, buying stablecoins is a way of cashing out. Furthermore, they are no longer dependent on exchanges that have to follow ‘Know Your Customer (KYC)’-regulations to offer their customers fiat currency.

These tantalizing examples show the unexplored ways in which we could exchange cryptocurrencies.

Money-Moving Metaphors

This friction of trades across the boundary is the origin of the clumsy *on-ramp - off-ramp* -metaphor, which resembles the outdated “Information Superhighway” that people used for describing the internet. In the future, we need a better metaphor to describe this process of small, closed circles of buyers and sellers becoming a larger closed circle. The right metaphor will inspire a better design for the user experience and is, therefore, an entrepreneurial opportunity. An example of a better metaphor is a scanner, a radar, for finding trades. The name of the decentralized exchange called “Radar Relay” refers to this. Other exchange companies have embraced this metaphor too. However, most of these exchanges only trade crypto-assets. An ideal exchange would allow anyone to sell any fiat currency for any crypto- asset, peer-to-peer. This exchange would have an automatic user interface. For example, the user only states a desired ratio or level for an asset, and the software handles the trades to reach these levels.

One type of potential trader of cryptocurrencies are merchants, who have currency as working capital. Merchants accept money from their customers and have to use part of that money to pay their suppliers. For now, at least a portion of these suppliers will want to receive fiat currency. Therefore, the desired ratio of crypto-assets and fiat currency differs per merchant. Furthermore, some customers want to hold a portion of their savings in cryptocurrency. While transacting for goods and services, merchants and customers have a growing need to rebalance their holdings.

Ongoing trade requires a rebalancing of fiat-versus-crypto working capital for the merchant and fiat-versus-crypto savings for the customer. In a fiat-only economy, or a crypto-only economy, this rebalancing does not happen, because no fiat-to-crypto transactions are necessary to rebalance; the balance is always perfect.

Conversely, in an economy that is part fiat, part crypto, people need to rebalance. As long as the economy has fiat and cryptocurrency, we need exchanges and stablecoins to handle the friction of rebalancing: that is why exchange services and stablecoins are in high demand. The exchanges, over-the-counter traders, and market makers have been the most profitable crypto companies so far. In a future in which most economic ecosystems will only use crypto-assets, exchanges will earn smaller margins.

To boost liquidity across the board and drive adoption, we need customers and merchants to make and accept payments in cryptocurrencies and other crypto-assets. The more holders, accepters, and payers a currency has, the more sense it makes to hold, take, and spend that currency. For any money to succeed, it must be liquid. Fiat currency becomes liquid because of the economic force of the nation-state that issues it. The more people exchange goods and other currencies for a nation's currency, the easier it is to find someone who will accept that currency as payment. A network of traders develops for exchanging large amounts of this currency so that no single trade of this currency can move the price; the market is so large that it can absorb shocks in supply and demand. In contrast, smaller markets, such as for an individual stock, are sensitive to changes in demand and supply. A portion of buyers and sellers can make large trades of that stock, which moves the price up or down more quickly than fiat currency.

The liquidity of fiat currency is a desirable quality: the more liquid an asset, the more stable the price. People who save for the short term hold a cash position to prepare for unknown circumstances in their company or personal life. Cash has the function of storing value when your future is uncertain; you get ready for risk and uncertainty by holding cash. People and companies that hold money want the value of the currency to be as stable as possible.

The potential of Bitcoin and any other cryptocurrency comes from its technical features, which *could* transcend the status of a commodity and become money. The technical features give the digital commodity advantages that motivate the pockets of buyers to hold and spend it as

money. As the isolated pockets connect into a closed-loop, they realize more of that monetary potential. The mechanisms that transform a commodity into money are visible as network effects. According to Investopedia:

The network effect is a phenomenon whereby increased numbers of people or participants improve the value of a good or service. The internet is an example of a network effect. Initially, there were few internet users since it was of little importance to anyone outside of the military and some research scientists. However, as more users gained access to the internet, they produced more content, information, and services. The development and improvement of websites attracted more users to connect and do business with each other. As the internet experienced increases in traffic, it offered more value, leading to a network effect.

A commodity can only become a currency when a large number of people agree that it has value. The usefulness as money can grow beyond the original value as a commodity product. Then, the commodity's price has two parts: the value as a commodity and the value as a monetary unit. The features and advantages of Bitcoin make its network effects hard to beat. Think about how easy or hard you find it to buy or sell a given crypto-asset. Compared to altcoins, it is quite easy to find someone to trade bitcoin, because there are so many more people who agree that bitcoin is useful. This recursive logic results in an iron dominance for the first cryptocurrency.

Other crypto-assets can try to challenge Bitcoin's dominance with features that are so attractive that they can overcome Bitcoin's network effects. Bitcoin does not keep its dominance on autopilot. Up-and-coming cryptocurrencies challenge Bitcoin all the time, so it has to stay better than the competition or become a relic. The digital scarcity does not only refer to the individual BTC tokens but also to the strength of the network effect itself. Establishing a new currency is so complicated that the first cryptocurrency that is good enough, BTC, is also scarce because it is accepted widely. Scarcity is not just limited supply; the market has to demand the asset more than it has demand for other assets that resemble it. It seems that the advantage of Bitcoin is substantial because the network effects of money are even more robust than those of, for example, audio

carriers such as compact discs and tapes, or a piece of software-as-a-service. Although you could come up with a system that is better than Bitcoin in some dimension, this contender has to overcome all the advantages of existing liquidity, brand recognition, security, and community that the Bitcoin network has acquired. For the foreseeable future, the odds are against that.

Other cryptocurrencies can still succeed in the ecosystem by aiming at niches left open by Bitcoin. Then the success of these alternatives depends on the technical limitations of the dominant cryptocurrencies. For example, people have different opinions on scaling the network's capacity, or the trade-off between privacy and ease of auditing, or on the consensus algorithm. Each of these viewpoints has led people to launch more altcoins. Bitcoin gave a boost to the conversation about a digital currency, and many people have become excited about creating new kinds of money. The game of money and finance is now an open field in which free players can compete.

The Whys of Crypto-Assets

The companies that rely on a cryptocurrency do expand the network so that growth begets growth. The positive feedback loop can snowball into an avalanche when the incentives are aligned. When each of the players in the ecosystem wants what another has, trading volume grows. To understand the motives of these people and see what makes them move a crypto-asset around, we can look at them through the lens of a *buyer persona* : an idealized and simplified type of customer, with a story that explains their motives and incentives. Software companies often clarify their thinking about their market by brainstorming these buyer personas, giving them a name, and then imagine that this person is using their software. We will look at these buyer personas for crypto-assets because friction in software for crypto-assets is always about interacting with technology. Before we discuss the technology in the following chapters, we first have to know the types of people who will use the hardware and software.

At this stage, your customer is an innovator or early adopter. They want your crypto product to get an edge, and it has to do with their motives. We can understand the buyer persona of users of cryptocurrencies in the context of a financial or monetary goal. That is why we have to define these

everyday actions as they apply to crypto-asset: earning, saving, investing, speculating, and spending.

Earning and Spending

You can earn a crypto-asset when you sell a product, service, or labor to someone willing to part with that crypto-asset. In case the asset is liquid enough, with a somewhat stable price in fiat currency, it will be easy to overcome the friction in the trade. That is because you no longer have to buy this asset anymore with fiat currency. You will have to make a transaction with someone who owns it and is also open to selling. That person also has to want to become your customer, client, or employer. We can expect a future in which most earnings of crypto-assets are cryptocurrencies, as these have qualities that can make them the most liquid; they are designed to be earned in the first place. As of the time of writing in 2020, earning or spending cryptocurrencies makes the most sense in the following cases:

- **Location-independent companies** with founders and employees across the globe, living in and traveling to multiple jurisdictions. Inside the fiat banking system, paying these teams is an ordeal. Bookkeeping and remittance become expensive and time-consuming. You can pay everyone in cryptocurrency because team members can sell their cryptocurrency locally. That trade is more natural than to have the company change whatever currency they hold into the fiat currency of a specific team member's country of residence.
- **Companies that want to attract new customers** by accepting cryptocurrencies. Some holders look for opportunities to spend as a way to cash speculative gains. Direct spending of their holdings saves them the costs of selling it for fiat currency.
- **Owners of companies that want to hold crypto-assets** who have a high-margin product or service; they only have to sell a portion to pay their suppliers in fiat currency, and they hold on to the crypto-assets or change them for other crypto-assets.
- **Uncensorable payments and donations** for controversial projects, companies, and NGOs. When you raise funds for resistance fighting,

protests, or other activism, when you sell gray-area or black market medicine, the earning party has adversaries, such as nation-state actors, intelligence agencies, or nasty competitors. They want to stop the incoming transactions and are willing to bear some cost for that. In theory, that means that the strength of your blockchain has to match their force. In practice, any blockchain that is supporting a liquid cryptocurrency will have that strength.

- **Private black-market deals** for some illegal goods that you can sell online. This is an arms race between law enforcement agencies and black-market sellers. Some cryptocurrencies keep boosting privacy features, and law enforcement agencies analyze blockchains and web traffic to catch illegal trade.
- **Sensitive payments** for hosting services, pornography, gambling, or prostitution—any good or service you might want to keep private. Cryptocurrencies never show up on your bank or credit card statement. Also, you avoid being cut off by your fiat payment provider for consuming these products and services.
- **Tax planning and, potentially, avoidance** becomes easier if the earnings never arrive in a bank account. They sit there, at a pseudonymous address on the blockchain, waiting to be spent or laundered. In case owners are willing to take the risk of prosecution, they can take a chance at avoiding taxes. They have plausible deniability until they move that money to another place that is connected to their identity. Naturally, they will delay that connection to their identity as long as they can. In fiat systems, parties and amounts in transactions are known by the bank and, therefore, potentially by the tax authorities. Although I do not endorse tax avoidance, I have met people for whom cryptocurrency is a piece in their tax-planning puzzle.
- **e-Commerce in low-trust societies** is impossible with fiat currency. An online merchant cannot accept credit card payments because the odds are high that the credit card is stolen, and the credit card company will reverse the transactions. They can not accept PayPal because these transactions can be reversed, and refuse to accept the burden of proof that they have shipped the goods. The same limitations also apply to low-trust industries with a high chargeback

rate, such as gambling and pornography. In a low-trust relationship, online purchases are only possible for buyers who are willing to pay upfront, in a way that can not be reversed by the buyer or a third party. The seller has to get a guarantee that the payment is final before he can ship the goods. Bitcoin and other cryptocurrencies can guarantee that finality, empowering new e-commerce.

When earning cryptocurrency, one source of friction is your payment system. When you sell products through a webshop, you have to automate acceptance of the payment, billing, and bookkeeping. The software that handles these payments has to fit the needs of the trading parties. This market for payment solutions is fresh and dynamic because cryptocurrencies, and the people that trade them, both change quickly. There are a few significant players in this market, but there is no guarantee that they can keep their position. Software entrepreneurs with experience in financial technology can look for an opening to enter this growing market.

From a technical perspective, spending cryptocurrency is similar to earning. The friction happens between the sender and receiver, such as between wallets and exchanges, or between a Bitcoin debit card and a point-of-sale system. Besides, the spender also considers that some cryptocurrencies deflate, that is, they can become more scarce and thus valuable. This adds to the friction of spending because the asset also has utility as a store of value. You estimate the gains you can make by continuing to hold the asset as part of your savings. Then you weigh that against the benefits of the goods you plan to spend it on. When you spend fiat, such a consideration does not exist. With crypto-assets, saving and spending are more at odds than with fiat currency.

Saving, Investing, and Speculating

We hold an asset because we expect it to keep or boost our purchasing power in the future. Holders of crypto-assets, or affectionately called “HODLers,” are assuming others want to hold, earn, or spend that crypto-asset in the future because it will be the best option for unstoppable, secure, and convenient payment or investment.

The saver wants to deal with the unexpected or to buy a specific “high-ticket item” in the foreseeable future. You keep the money safe while it is accruing. If your savings are in fiat currency, the bank keeps it secure. But

now you have the option to rely on cryptography instead. You can secure savings in new ways. With personal cryptography in the mix, storing value can be safer and more natural for some people than in banks. We will discuss some of the details in the chapter on wallet technology.

Investors want to keep value over time and generate a return from it, such as when they buy a piece of real estate and rent it out. We can imagine that tokenized real estate becomes more natural to trade between people that have a wallet. Instead of owning one building, or a pooled investment in real estate, you can divide the property into many small pieces that become tradable between people with wallets.

Speculation on the future price of bitcoin and other top crypto-assets is the primary driver of the price. The concrete benefits of BTC, ETH, and other available assets are limited and not the cause of the current price level. To explain the price, we have to account for the future. Crypto-assets promise a set of future benefits, related to money and finance, on which you can speculate by holding. Stories about the future are various and unstable, with price moves driven by whimsical speculators. However, crypto-assets can already be a tool in your portfolio because they are a new asset class, and the price tends to move independently of other assets.

The bright future of crypto-assets can only become real with benefits for specific buyer personas and use cases. We can explore that space of possibilities by asking questions that start with a lack. For example:

- Ordinary BTC transactions are too costly for small payments.
- Wire transfers are too slow and expensive for retail payments.
- PayPal is not secure and final enough for receiving e-commerce payments as a merchant because they can and do freeze your account.
- Credit card companies charge the merchant a percentage per payment and finalize the payment after 60 days.
- TransferWise is not open enough for black market transactions.
- Credit card companies and PayPal will not process payments of the online purchase of some medicines.
- Credit card payments are not free (as in speech) enough for buying medicines online or controversial donations.

- Savings accounts and treasury bonds are not secure enough for long-term savings because of the risk of currency inflation or outright collapse of the debtor.
- Paper cash is unsuitable for payments over long distances.
- Informal payment networks are too slow or insecure because of vulnerability to a third party.

As money becomes a free and flexible medium, we can expect these problems to disappear. These containers do not limit an economy that relies on cryptocurrency. Money is money: why should an arbitrary medium or custodian for your wealth determine how you hold or spend it? Once you see the features of ideal money, you itch to transcend these limitations of particular media. We will explore the opportunities of ideal money in the next chapter.

The entrepreneur and investor Paul Graham discusses the source of startup ideas in his essay *How to Get Startup Ideas*. His suggested methods boil down to: “Live in the future, and work on what seems interesting.” Another way of saying that is: notice what is missing. Now that we have looked at how crypto-assets could be used, we review a few examples of these innovating buyer personas to see how crypto-assets enter their reality and what perspective they have on advanced forms of money. Entrepreneurs who start a company in the current crypto space will sell to at least one of these buyer personas. It is worthwhile to know to whom you are selling.

Early Adopters as Potential Customers

There are multiple reasons to care about cryptocurrencies, and each idea motivates a type of crypto-enthusiast. These different enthusiasts help to grow the network of a given crypto-asset. These reasons for caring about cryptocurrency often combine in a single person, with one or two motives at the forefront. These types of people show up at conferences and meetups to discuss cryptocurrencies and tokens in detail; they are often working professionally with crypto-assets or have invested a portion of their savings. In Everett Roger’s model of *Diffusion of Innovations* (1962), they are the *innovators* who are willing and able to take risks with new technology. As an entrepreneur, you might care about an innovator’s opinion. Can you engage their imagination and inspire them to act? That is a test your product must pass. They buy into applications, platforms, and crypto-assets that

move their crypto journey forward. You want them to approve of your product or service.

At the time of writing in 2020, cryptocurrencies have diffused to the next stage of diffusion, *early adopters* : opinion leaders who put their cryptocurrency where their mouth is. They drive the story forward to the *early majority* . When we add up all these journeys, we get a megatrend: the replacement of the fiat status quo with sound money, unstoppable code, crypto-assets, and decentralized finance.

All crypto enthusiasts I have met behave like one or more of these five buyer personas:

Playful Developers

A software developer builds systems. They may look serious in their studies and experimentation, but their work feels more like creative play. Solving problems is their entertainment. They like to tinker with hardware or software, and they often have a job or freelance career in technology, with the skills to develop software. They get paid to solve problems of design and architecture to make the systems work. They are always looking for more tools to play with and to get things done. The goal is to build something exciting and new. When a developer gets interested in blockchain technology, they focus on the possibilities of the tools. They look at the code to see how they can use it to build a system that supports a new type of transaction.

The drawback of a developer's tool obsession is the risk of digging into a technical rabbit hole, where they miss out on the big picture. When spending too much time on the details in the code, their wrong-headed ideas show up in the metaphors they use to describe a blockchain. They focus on the trees and do not see the forest. For example, I have met a blockchain developer who compared the feature of 'immutability' to "a house of cards," which is wrong. He was focusing on how a hash function depends on the input, leading him to choose an image for fragility. However, the quality of immutability requires an image that evokes something heavy and solid, such as "layers in a glacier," suitable to describe the immutability of the transactions. Another example that surprised me was meeting professional Ethereum developers who had never thought about the origins of money. They understood how particular crypto-assets worked in great

detail but did not appreciate the greater context in which these assets work. Specialization has opportunity costs.

Playful developers care about Bitcoin and other blockchain technology because it defines a field of possibilities to build new trinkets and systems. They can resemble the teenagers in the 90s who switched from print board electronics to computer programming because any new thing is cool. As the hands-on builders of the crypto sphere, we find developers in the role of supplier, founder, or employee. Much of the resources they use are open source or low cost. As early adopters, they are likely to become a customer of one of the other types.

Adventurous Entrepreneurs

Starting a company is ambitious, and the type of person that starts a startup that uses bitcoin, blockchain, or crypto-asset tends to have a ‘larger than life’—personality and a vision for the future. Most entrepreneurs are generalists who can see the big picture. Some are specialists in several areas relevant to their industry. Furthermore, they have the magnetism and leadership that draws a team around them. Like Steve Jobs or Elon Musk, their passion and energy generate a “reality distortion field” that supports return-on-effort for a group of specialists with a common goal.

Bitcoin, Ethereum, and maybe other blockchains will empower entrepreneurs to take out the middleman. The entrepreneur gets excited by opening a market and disrupting the incumbent monopoly power that dominates the field. Entrepreneurs of generation X and Y know disruption in their gut because they have seen the internet grow up from a toy into a global utility. Now they want to harness the disruptive force of immutable blockchains to disintermediate incumbents who own the traditional financial system.

The entrepreneur’s challenge is to pick the right project. It is tempting to take the dream too far by plastering a blockchain on top of an arbitrary industry, and pray that investors, potential team members, and customers see the light. Such a rushed blockchain startup is a bad gamble. A good startup idea comes from understanding the problems in your industry and knowing your tools. Furthermore, you have to understand your customer’s situation first. It is reckless to assume that “the blockchain” can solve, for example, the infrastructure inertia of your favorite supply chain. If you are going to “bank the unbanked,” you better know exactly how your startup

moves money better than the informal network of relationships currently serving that demographic.

Blockchain entrepreneurs can protect themselves against a lousy startup idea by slowing down. They can study the basics and recent history of Bitcoin and Ethereum to get familiar with the ways a project can fail. The quality test for your blockchain startup idea is a trial-by-fire: can you pitch your idea to the early Bitcoiners, often referred to as “original gangsters,” and answer their pointed questions? The best places for these discussions are Bitcoin meetups and cocktail parties at blockchain conferences. Or, if you have a good web presence, ask your followers. If you and your idea can hold their own in a debate, your startup is in good shape. However, if a crypto veteran can incinerate the plan with one magical question, you are likely better off ejecting with a burned ego than burning investor money.

As I suggested in the introduction, entrepreneurs are my primary audience. It seems that this type of crypto enthusiast will have the most influence on the diffusion of crypto-assets in society. They will have to take the risks that push this technology into the mainstream.

If entrepreneurs are your customers, aim to connect them to crypto investors. In technology hubs like San Francisco, Hong Kong, Singapore, etc., there already exists a lively ecosystem for connecting the capital to labor. We can view the ICO boom that peaked in 2017 as the first step to decentralizing that ecosystem. Retail investors now have options similar to those of angel investors and venture capitalists (VCs), and they need tools to make sense of available deals. Also, entrepreneurs need the tools to organize a token sale and process payments.

Although the market for ICOs has its growing pains, the globalization and democratization of startup funding is too large a trend to ignore. Issuing a token to a global market of retail investors could potentially be faster than selling to an elite of angel investors and VCs. Although much capital is available in technology hubs, that amount is small compared to what is available worldwide. Venture funding at a large scale and top speed have a place in the future of crypto-assets. As an entrepreneur, you can prepare for fast and easy financing of the best ideas.

Activists, Anarchists, and Libertarians

A portion of crypto enthusiasts pursues individual liberty. They have political motivations and see their crypto-assets as tools of activism to fight

a system they consider unjust. Anarchists and libertarians have minority views of property rights, and they resist some of the mainstream methods for acquiring and defending material wealth. Some oppose intellectual property rights. Some oppose specific taxes or taxation in general. Others disagree with the government's monopoly on violence and law.

The lovers of liberty care about the political effects that a given crypto-asset can support. A new balance of power between centralized government and individuals could mean economic transition toward a better society. Activists try to liquidate unjust power. For privacy advocates, anarchists, and libertarians, Bitcoin gave rise to a new hope. Instead of a fruitless debate about government finance and fiat currency, they can create, buy, and spend their preferred money. The existence of crypto-assets enables them to exit from the financial system that disagrees with their values.

Anarchists and activists will use software that protects them from surveillance and censorship. They want private internet and pseudonymous identities, to support freedom of speech, freedom of association, and freedom of transaction. The more accessible these tools are, the more mainstream they become. For example, many observers expect the rise of a decentralized social network that can replace Facebook and Twitter, which are losing their reputation to scandals of privacy and censorship. For now, centralized social networks are "better" because they are quick and easy. As soon there are social networks on which you are private and uncensorable without a loss of convenience, the centralized social networks are disrupted. Several startups have built identity systems and protocols that adapt to people who care about privacy and anonymity. Pseudonymous and anonymous identities and rating systems can be a part of the future of crypto-assets. All we need is people to build them.

Traders

Traders aim to buy low and sell high. Successful traders depend on a decisive information advantage; you know more than the other players in the market. As a trader, you see a future reality that is yet unknown to your opponents. To profit consistently, you need understanding and experience to develop that advantage. Your competition is using the best data and algorithms available, and you need something that gives you an edge.

Trading has a steep learning curve. Learning to trade is like becoming a medical professional; Doing the first-aid course is not the same as going to

medical school. Studying massage therapy for four years is not the same as becoming a neurosurgeon in fifteen years. Medical sophistication is a deep game, and “day trading” and arbitrage are similar to medical skills in depth and complexity. However, unlike medical expertise, the risks and rewards of trading are varied. Most people who enter the market will lose their money to the most advanced professionals. We classify traders based on their time horizon:

Arbitrageurs profit from price differences between marketplaces. They use trading bots on exchanges or deal in crypto-assets “over the counter” (OTC) with buyers and sellers in their professional network. In the ideal case, arbitrage has no risk because you buy and sell at the same time, profiting from the price difference between the two markets. These windows of opportunity open and close quickly.

Day traders sit behind their screen full time, analyzing the market with charts and algorithms. They often use bots that run optimized algorithms to find and execute the best trades. Day trading requires nerves of steel and the ability to sit still. Many crypto traders had previous careers in online poker or traded fiat currencies with leverage (FOREX).

Position traders stay in or out of the market for several weeks or months to ride a trend. You can do this with a smaller time commitment as it is not as labor-intensive as the other types of trading. But it is harder to profit, as you are unlikely to have an information advantage.

HODLers invest and plan never to sell; they bet that the asset they hold will replace fiat currency for saving, paying, and for pricing products, services, and investments. HODLing requires a long time horizon and a degree of understanding and faith in the held crypto-asset. This strategy is limited to the assets with the highest market cap, such as BTC, ETH, Ripple, and top privacy currencies.

As a group, traders are looking for tools and data that help them analyze the market and connect them to new opportunities. You can attract them as customers by building a superior exchange, platform, or source of data.

Industry service providers

The blockchain industry has much building and educating to do. The technology is still scrappy, opaque, and confusing. Further adoption of crypto-assets by new markets requires a smooth user experience and storytelling. There is work to do, and marketers, designers, and educators have started offering services to startups that develop crypto-related products and services. With the right timing, they can be on the right side of history.

As a service provider, you gain an advantage by working in a growing industry. On the one hand, it is riskier because there are fewer clients, and the marketplace is immature. On the other hand, the service provider can establish a first-mover advantage and settle in a comfortable niche before the masses show up. These new service providers have a good sense of timing and momentum, seeing that Bitcoin, blockchain, and crypto-assets are the next big thing. They help entrepreneurs and developers who build the ecosystem. These types tend to be designers, educators, and marketers.

The entrance of these supporting service providers suggests the innovators are becoming connected to early adopters, who are more numerous. At first, the typical crypto enthusiasts were of the other four types, such as developers, traders, and anarchists. Suddenly, there were waves of people showing up with a marketing and communications background. They are motivated by the opportunity to apply their skills in exciting ventures—the crypto space benefits from this pool of talent.

CHAPTER 4

Ideal Money

PEOPLE ADOPT CRYPTO-ASSETS that have a specific function in their economic life, such as transacting, saving, speculating, or investing. In the previous chapter, we saw that crypto-assets that fulfill those financial and commercial functions catalyze a network effect, giving birth to a self-reinforcing cycle. In essence, the more people use the asset, the more liquid it becomes, which in turn boosts its usefulness and value, which leads to the price rising. This high price allows the network to absorb more transactions of higher value. Rising liquidity makes the asset more stable, which draws new users, which makes the asset more liquid. The benefit of this feedback loop makes the cryptocurrency or asset a growing success.

The features of a crypto-asset that catalyze a network effect fall into two broad categories:

A) Financial choice —We can judge these as value propositions with features and benefits to a specific buyer persona. The users of the product or service get choices or options that are black and white; they make the transaction possible or not. The entrepreneurs who can invent these features apply the traditional insights into startups to the domain of finance. Most startup entrepreneurs are familiar with the finance industry as it was *before* Bitcoin, and therefore overlook money, financial assets, and transactions as areas where they can win. But now, entrepreneurs merely have to turn their heads to see the opportunities. Strict regulation of international banking makes many transfers slow or impossible. Any substantial sum becomes precarious. Regulators force international bankers to act as third-rate cops that ask annoying questions. From the perspective of the client, it looks like petty abuse of power. They can use cryptocurrency to evade such customer experiences.

Another example of giving people financial choices: Crypto-assets that relieve a person's tax burden would be in high demand. Perhaps you can make people's income streams passive, moving them from income-taxed to

capital gains tax. There are likely many opportunities for regulatory arbitrage of tax law. Wealthy people already do this, with the help of costly attorneys, who are middlemen of traditional law. In the future, entrepreneurs can use smart contract platforms to provide those services to anyone who feels that they pay too much tax.

B) Low friction —One factor in the growth of the network effect is friction in the user experience of a crypto-asset. The resistance depends on, for example:

- How quickly can a user get on board?
- Does the user need technical knowledge?
- What is the learning curve for the typical user?
- Does the software evoke the right metaphors to support the mental model of the user?

To apply an idea by consultant and educator Peter Drucker, crypto-assets that fulfill the benefit of transacting and holding are *effective* : they do the right thing, while crypto-assets that make these transactions quick and easy are *efficient* : they do ‘the thing’ right. We see that a specific crypto-asset first proves its effectiveness, and then becomes more efficient. In the case of Bitcoin, users first needed to understand the notion of a digital, scarce, unstoppable currency. A critical mass of enthusiasts had to want it badly, and then entrepreneurs could profit from building exchanges and wallets that made using Bitcoin fast and easy. Altcoins showed a similar development, where the more successful crypto-assets got support from exchanges and wallets. The network effect of a cryptocurrency depends on the usefulness first: it has to empower a new type of transaction effectively. The currency or asset needs a reason to exist to get the feedback loop started. Once the users can fulfill their transaction needs in this new way, they can care about its efficiency. The transaction is a must-have; its speed and comfort are nice-to-haves.

Many of the founders of altcoins confuse monetary effectiveness with payment efficiency. They compare the speed and ease-of-use of Bitcoin and Ethereum to centralized networks such as regular credit card payments or mobile applications of banks. These projects try to launch a cryptocurrency that competes in efficiency with fiat payment networks, without introducing

the new type of transaction. However, there is no point in printing money that has no reason for existing. For now, digital fiat is doing an excellent job handling convenient payments. The novel use case, which unlocks new economic value, is a necessary condition for success.

Building the Higher Layers

One promising technology for making transactions on a blockchain faster and cheaper is a second protocol layer, on top of the base network. The protocol allows nodes to open channels between them that keep the transactions open until they close the channel. Then only the final balance is settled on the base layer. That will remove the need for maintaining a public and distributed record for small transactions. Protocols such as Lightning Network can ‘modulate’ the features of a bitcoin. For example, the users do not need to pay for hard guarantees if they make a small payment. A retail purchase such as coffee is not significant enough to record on the base layer of Bitcoin or Ethereum. The system should aggregate these on a higher tier. Also, you do not have to settle the balance daily. Many companies, such as retailers and subscription services, have customers who come back regularly, so an open payment channel is more efficient.

The efficiency of the second layer protocol is like the difference between a paper contract, which is fast and cheap, and a court case, which is slow and expensive. A well-defined deal prevents the court case. The transacting parties only need judges for the rare and unlikely event not covered by the contract. Likewise, the smart contract on the second layer will usually work fine, and only if a dispute does occur, does either party need the forced on-chain transaction to settle the balance.

The second layer will free the media of money in the same way that internet protocols free information and communication from their traditional carriers. In the old days, we had channels and “media” to hold and transmit data, and publishers had to squeeze the message into the medium’s limited economics.

Some examples:

- You took pictures in sets of 36 and could see them, after “development,” a week later.
- TV and radio are uniform and passive.

- A telephone is for one-to-one, and talking to people further away was more expensive.
- The newspaper informs the opinions within a nation.
- Compact discs (CDs) had two or three good songs, but you pay for all of them.
- Private video recordings were scarce.

With tablets, communication became easy, cheap, and frictionless. The trend of smartphones, tablets, and cloud computing finished the move away from specific media. Information flows lightly one-to-one, one-to-many, many-to-many, and many-to-one.

How Software Frees Money

As discussed in chapters 1 and 2, there was one particular kind of “message” that was a lot harder to free from its media: money. That is because monetary media have to be scarce, portable, and trustworthy. Up until 2008, no one knew how to make all three of those properties digital, so money could not yet become as light as information. Money had been digitized, but only at the surface. You pay online with your credit card and PayPal, you do online banking, and financial institutions have been using information technology since the IBM mainframe. This step of digitization only had software liberate one of the properties of money: portability. Unlike cash, gold, or diamonds, a bank can move fiat money around like information, like the “messages” of 20th-century media. Digitally transporting fiat money was the first step. However, the other two properties, trustworthiness and scarcity, still depended on the physical and corporate nature of authority. The bank had to guarantee the accuracy of your deposit; you had to trust them to keep the books correctly. Furthermore, we had to trust the central bank to create the exact right amount of money and to set interest rates. We had to trust the government to impose the exact right kind of regulation on these banks. Finally, the money medium most resistant to being replaced by the software was cash—the banknotes and coins for paying the bartender and cab driver.

Since Bitcoin, the trust in the rarity and desirability of the money is being freed up by software too. That includes the function of fiat cash, although the belief in and value of cryptocurrency as cash is still in development. Cash payments can become digital when transactions are

practically free. The fees should be much lower than the opportunity cost of going to an ATM or ADM. Wherever Lightning Network technology is adopted, fully digital money wins. In one of his talks, crypto-educator Andreas Antonopoulos calls it “streaming money.” In this book, I refer to it as *ideal money*.

Who would want to use fully digital money? On the one hand, we know that in many developing nations, people whose rulers corrupted the financial system are ready to switch to cryptocurrencies. On the other hand, developed countries have the luxury of trusting monetary authorities that appear to be good stewards of the fiat currency. Most people will accept the national fiat cash as payment and trust their bank to keep it safe. Under which circumstances would stable liberal democracies also adopt cryptocurrency? Only when crypto-assets become as light as information and free of any medium. When your money is invisible to anyone but you and whom you are paying. Money that moves like information but is as hard to create as gold. Money that can not be stopped or confiscated.

When paying with cryptocurrency is more comfortable and faster than anything else, and it is sound money that we can stream anywhere, anytime, it is up to entrepreneurs and developers to make new types of payments available. It is like predicting video conferencing and e-books in 1994: We can see ‘the internet of money’ coming, but the ‘Facebook of finance’ or the ‘Google of contracts’ are probably not imaginable from where we are now. We can only intuit the possible and prepare for the opportunities. Here are some examples from our current vantage point.

1. Pay Subscriptions Per Second

Present situation: A monthly or yearly subscription, like the newspaper, gym membership, salary, club membership, and tuition is paid this way because the bank transfers the money in chunks. We calculate the cost relative to our income, budget, or credit card balance. The seller also benefits from the subscription model: newspapers or gyms recover the onboarding cost when you commit to a 3-month minimum.

Future situation: Your gym membership is one smart contract that defines a stream of money from your wallet to the gym. You can cancel at any time.

That means you only have to make two decisions: sign up, use the gym, and close the stream when you no longer want to use the gym. You do have to think a bit before you start because the flow of money is not constant: it begins at a high rate per minute, let's say the equivalent of a \$20-day pass. To your wallet, that means about 0.014 cents per minute. Over, say, ten days, the stream slows down to the "1-month membership", \$90 per month, or \$3 per day, or 0.002 cents per minute. The rate per minute continues to drop to about 0.001 cents by the time your membership lasts for six months. It will settle there forever until you end the contract.

This type of contract also changes the power dynamic with clients who keep service providers on a retainer. The convention of transferring the money each month makes this deal 'unfair' by default. Either the client has not yet gotten what they paid for, or the service provider is taking on debtors risk. These business relationships require trust. However, now you can trust the contract that controls the money stream, so a broader range of client-provider relationships opens up.

Note that the mental effort is little. People already understand memberships; they know that the service company wants to reward their loyalty. Instead of a table of options, you see the graph of money per second going down with the length of the contract. That way, a steep descending graph might become a selling point. I can imagine that we will get used to those graphs, as we have gotten used to smartphone specifications and road signs. Soon, we could have a sense of how a smart contract for a money stream is supposed to look. Then we will have a new vocabulary for money.

2. New Metered Commodities

Present situation: Water, electricity, gas, and taxi rides are charged per unit. There are two variables: there is the 'starting fee' and then a constant rate for a resource consumed and time of use. Taking a cab is different from place to place: sometimes you agree on a fixed price, sometimes you rely on the meter to be fair about it. With applications such as Uber and Lyft, you can send your money to the driver according to that agreement.

Future situation: The contract can become even smarter by including GPS data of the route—that protects the customer from 'confused' cab drivers who take a detour. As always, incentives are everything.

Products and services that we now pay for with a subscription plan can become metered commodities. Internet service providers can charge more for fast bits and less for slow ones. Users can tweak their preferences. By setting an upper limit for prices, the user can overcome the hurdle of mental effort.

3. Rates Charged by Consultants and Coaches

If an appointment is considered a strict agreement between the participants, then the wallet can talk to a shared calendar application and GPS locations. The people in the meeting can no longer make excuses about lateness and cancelations. Jobs that are not easy to define in terms of time and deliverables can have contracts that start the meter whenever the client and consultant are together at a designated GPS location. Any day this happens, the money stream includes a minimum amount for travel time and cost. Alternatively, a fixed amount per minute retainer keeps the client caring about moving their part of the work to the finish line.

4. Smart Healthcare Contracts

People that have a high-grade healthcare plan already know what this is like: you see the doctor, they treat your illness, and you walk out the door. That is because health insurance is already a kind of proto-smart contract. Insurance companies make deals with hospitals, and payments are regulated. Smart contracts of ideal money are likely to take over those arrangements.

Present situation: Healthcare has many perverse incentives. Insurance companies and a portion of healthcare professionals view their patient's illnesses like lousy weather: it just happens. They don't spend too long looking for root causes. Instead, they manage the symptoms and apply the known cure. The patients care about their premiums but not the medical bills. The insurance companies push for discounts for buying treatments in bulk, but do not worry about the hospital staff's administrative overhead. The medical staff doesn't mind high prices for treatment; they want to treat patients quickly with minimal hassle. The hospital management and shareholders also have their own set of costs and benefits. All these conflicting incentives explain the challenges of the healthcare system in many countries.

Future situation: Behavioral data from patients can fix the incentives for each of the stakeholders. Patients can sell their behavior data, piece by piece, to any of the other three stakeholders in their health. Big data from masses of patients can lower their premium in real-time and raises the collective intelligence of the system. Smart contracts based on paid data take us into a future of fewer, happier patients, easy-going doctors and nurses, provably fair insurance contracts, and wealthy hospital owners.

5. Right-Priced Flights and Rides

Present situation: The fixed price of a ticket for public transport is the outcome of in-house accountants analyzing the cost structure and setting a price for some time. That is smart enough for the standard ride. However, some trips are not standard. The weather causes flight delays. The railroad track is blocked for a while. The bus is rerouted because of road construction. In such cases travelers have to accept their misfortune and will regret their purchase. The only thing they can do is to choose another means of transportation next time.

Future situation: What if the traveler can get their money back in proportion to the delay? Insurance company AXA, through the brand name Fizzy, is already using the Ethereum blockchain to automate insurance for late flights. The contract is still crude: if the plane was two hours late, you get a refund. However, we can expect a rising level of granularity of the data for right-pricing tickets.

All prices per unit can become ideal money so that the figure in your wallet application will move at various speeds in either direction.

6. Paying for Attention

The proven way to monetize attention is through advertising. Online advertising has become granular and efficient already. With ideal money, new transactions become possible between content creators, audiences, and advertisers. The exchanges can start flowing the other way: advertisers can pay their target audience to review an offer. Moreover, viewers can pay for content directly so that they do not have to look at ads. The companies Brave, Steemit, and Earn have applied this model. As ideal money becomes the norm, we are likely to find other ways of pricing attention. For example,

some companies let you pay for razors on a subscription plan. That saves you the effort to remember when to replace it by moving the product purchase into a subscription ‘contract.’ With ideal money, innovations such as these can turn up elsewhere, at larger and smaller scales.

Smart contracts with ideal money become a part of our lives so that we can outsource more financial decisions to computer programs. These algorithms will become more intelligent as a result. These data-fueled AIs will give us the luxury of not having to think about money that much.

7. Paying for Content

Present situation: Content no longer needs a specific medium. Digital publishing has near-zero marginal cost; that is, the cost of reproducing the story or information has nothing to do with the number of people viewing it. That means you can no longer charge for delivering your content to your audience; the free reproduction makes the material so abundant that any charging for content becomes a race to the bottom. One solution is to charge for convenience and curation. With entertainment, such as novels on Amazon and shows on Netflix, the seller charges so little that most people do not bother with pirating. With non-fiction information, online companies sell the content as part of a journey with a new identity and a lifestyle. Examples of this are Mindvalley.com and Masterclass.com. Paying for the content is pitched as a logical step in the journey. When sold at the right price, enough people will pay for that journey instead of pirating.

Future situation: Content creators can make a smart contract that allows them to gamble on their audience. For example, when you empower your readers/watchers/listeners to get business success, you get a portion of their revenue. You become a mini-Amazon or Shopify. Ideal money will make it so easy to pay tribute to your teachers and gurus that it can replace traditional education and credentials.

The progress of the Lightning Network implies that we can expect use cases for cryptocurrency to become available gradually, as fees fall. For all small transactions, we can now assume a future where most payments are almost free. The only remaining bottleneck is the size and growth of the network of users, which depends directly on usability.

These examples show that payments will become granular, so that ideal money enables new value propositions by right-sizing and right-timing any

amount. Money has finally entered the age of information.

CHAPTER 5

Wallet Design

IN PREVIOUS CHAPTERS, we discussed the removal of friction in handling money; one way to make crypto-assets easier to use is better wallet applications. Wallets are an entrepreneurial opportunity because the sellers can choose from proven business models: you a) support the trade of assets by connecting to an exchange, b) charge for the hardware of the wallet, or c) support secure backup for a monthly fee. Some wallet startups are experimenting with new business models, but it is too soon to tell if they will work. That is why this chapter is just about ease of use and security, with the assumption that if you build a safe and easy wallet, the paying users will come.

Better user experience helps to bring crypto-assets to a larger group of adopters. The interface to crypto-assets, a “wallet,” lets you send, receive, and hold crypto-assets. However, at this point, only tech-savvy people, the “innovators,” use them with confidence. Even the best wallets are, at times, awkward to use. In the coming years, wallet developers have to tinker until grandma can give her grandchild a CryptoKitty for their birthday.

I have mentioned the point at which I felt I could buy Bitcoin with confidence. I had attempted to buy Bitcoin much earlier but gave up because of the friction and risks. The wallet technology and the market penetration of exchanges had to reach a certain level of ease before I became a buyer. My case is one example among many. No matter your technical expertise, motivation, and talent, at some point buying and owning Bitcoin was over your head. It is not just about your access to a market; you also have to be capable of securing the asset. This also applies to the Bitcoiners who could get in very early by mining on a laptop. I have heard and read many accounts of screwups and theft. The following situations are typical:

- Some early Bitcoiners mined many coins but did not keep a backup. Corrupted hard drives or partners throwing out “old computers” have cost them dearly.

- Some people secured their crypto-assets with overly complicated methods, forgot a critical step, and could not recover their property.
- Some kept their security setup simple but forgot their password after an accident that involved a blow to the head, or forgot over time.
- After Bitcoin became more mainstream, thieves started targeting known holders with hacking, break-ins, or kidnapping combined with threats of violence.
- Hackers use phishing attacks, trojan horses, and infected websites to enter wallets on people's personal computers. I witnessed one such case up close. The victim had \$30,000 worth of crypto-assets on a laptop dedicated to this purpose, stored in multiple wallet applications. He had installed a Monero wallet that turned out to be malicious. After getting access to the computer, the attacker drained it and all his other wallets as well. The victim had a professional background in information technology, but there was nothing he could do to stop the theft after he downloaded that compromised wallet.

Tragic events such as these are symptoms of the innovator stage of crypto-assets. We can look to a future where such problems are as rare as errors in a bank statement. In recent years, wallet designers have already addressed several of these security issues—we will cover this below.

Remove Friction

To generate design ideas for wallets, we can compare types of money and payment technologies in terms of friction. Each payment method has medium-specific limitations, and wallet startups can tackle each of them as challenges to solve. A financial transaction requires the user to undertake a series of actions inside a user interface in order to reach their goal. We distinguish these steps as a) user onboarding and b) repeated use. For example, in some countries and for some people, applying for a credit card has a two-month lead time and represents a five-hour workload. In other countries, you get a functional credit card in the mail without even asking for it, provided that you have a good credit score. Using a credit card in daily life is standard, but credit cards are far from ideal money, as discussed in the previous chapter. This example shows that technology can have friction at different levels and points of contact, depending on the user's

characteristics. Designers can watch users and use empathy to find the points of friction for their target market.

In the long term, we can expect each of the methods of payments to get bundled into a single wallet technology that supports crypto-assets. This technology will be built by new companies or open-source groups, *not* by banks. A complete wallet application will hold fiat currency as well, but keeping regular electronic fiat currency requires a banking license, while banks are unsuited to support crypto-assets. That is why the fiat currency in this type of wallet will likely be ‘stablecoins:’ fiat currency, usually dollars, represented by tokens on a blockchain. That solves some of the problems of the on-ramp and off-ramp, which we discussed in Chapter 3. A complete wallet will hold at least all the well-known and liquid crypto-assets, and remove friction in fiat currency/cryptocurrency trades.

Crypto-assets could develop similarly to the internet: from strange and risky experiment to safe and effortless utility. The innovators who built the internet could only start disrupting telephone and publishing companies when they began serving the mainstream. The first step to mainstream adoption was the development of internet browsers. Subsequent waves of adoption relied on webmail, fast-enough video, BitTorrent, social networks, and smartphones.

Mainstream adoption of crypto-assets depends on a wallet design that offers new metaphors for spending, earning, saving, investing, and speculation. We have to adapt the money interface to people’s existing ideas of how money works. For crypto-assets, the percentage of market penetration is still low; we are in the innovator stage. If you understand it well enough to use it, you are an innovator—like users of the internet in 1995. Every innovator has a unique opportunity: you could accept responsibility for moving the field of crypto-assets forward by reaching early adopters. When we zoom out, the replacement of ordinary paper and digital money by crypto-assets seems inevitable, but we still have to build that future. That paradox points to the work of an innovator; as an entrepreneur, you need a sense of destiny combined with a sense of urgency.

Under the Hood

In available technology, complications are hidden “under the hood.” In the early stages of development, you see and hear all the gears. Then the gears

move into the background until they disappear, leaving pure functionality for the user to enjoy. The perfect sound system is just sound. The ideal TV is just a window into somewhere else. Small children do not know that gasoline is flammable and that cars run on explosions, like dangerous fireworks, enclosed in a safe little space under the hood. The internet used to run on phone networks optimized for human voices, which is why your computer made strange noises when it was dialing into the system to go online. Now the internet has become a general abstraction, a utility like water and electricity. In time, people will be online without having heard of WiFi, Bluetooth, and 5G. Whenever the connection breaks, they will be surprised that they can not study, work, or talk to their friends.

We have to push the usability of crypto-assets onto a similar path. The history of technological development shows that the ideas we need are right under our noses. However, homo sapiens tend to be myopic, locked into the status quo of our surroundings and routines. Hunter-gatherer conditioning resists change because change often means danger. Interface designers and security experts have to transcend that ancient instinct and use their imagination to break free of dated models and traditions.

The Design Space

Crypto-wallets are tools with a user interface with many imaginable designs. Tools vary in complexity. For example, anyone can use a pair of scissors, while it takes training to fly a plane. That illustrates the vast *design space* that blockchain interface designers have to explore. The possibilities for wallet designs are endless, and only a small portion of those will function in practice. Some designs will be found quickly, while others will take years to discover. In the meantime, we can draw inspiration from the maturation of internet software as a historical parallel.

The first distinct advantage that the internet offered was e-mail: users send messages to someone on their contact list, they have an inbox, an outbox, etc. You could imagine that the next step is: let users connect to strangers and send them messages. On the surface, it appears an equally simple design challenge. You collect a group of strangers on a website and give them each an inbox and a profile. A dating site, for example. However, it took about fifteen years for dating sites to develop from ‘terrible’ into ‘good enough.’ Other examples that illustrate the size of the design space are Upwork and LinkedIn. Both platforms match employers with

employees or freelancers, yet their target audiences and visions are different. Websites and applications specific to online dating or job matching are now somewhere in the ‘early majority’ stage. This historical parallel suggests that Bitcoin and other blockchain technologies have many details to tweak until we have a quality experience to offer mainstream users. To sketch the design space for crypto-wallets, let’s have a look at the challenges and tradeoffs that we face now, in this early stage.

The Risks of Holding Private Keys

Crypto-assets are a type of information; they are scarce but not made from material substances. This lightness of valuable information generates specific risks for wallet users, which designers have to manage. Crypto-assets are stored on a blockchain with a public key, at an ‘address’ secured with a private key. Public and private keys are big numbers, represented as strings of digits and characters. The wallet’s keys are the most crucial element; whoever controls the private keys controls the crypto-assets, so the private keys to your crypto-assets should never leave your device.

On the one hand, the keys are what the bad guys want to steal, and on the other hand, you must always have a backup of the keys in case you lose or damage the hardware device. Although keys are information, the risk is physical because a criminal could coerce the owner into surrendering the key to the crypto-assets. This security tradeoff is the number one concern of wallet functionality.

Wallets should be easy to use, but the designer faces a dilemma: convenience implies risk. You can boost security by removing risk, which means the users pay the price in terms of money or hassle.

The dangers of theft or loss of crypto-assets resemble a substance that is also liquid and volatile: nitroglycerine. One has to handle this oily explosive with care because if it vibrates, it blows up. Chemists learned to stabilize nitroglycerine with absorbents, giving you dynamite, a tool for blasting tunnels through mountains or demolishing buildings. Today, crypto-assets are still a dangerous substance. Security experts and interface designers have yet to discover the ‘absorbents’ of crypto-assets—the methods that protect non-technical users from theft and loss.

Part of the solution to secure crypto-assets can be found in history and nature. Before liquid assets could be information, they were always physical objects, such as jewelry, gold bars, silver coins, and gemstones—

assets you must either hide or defend. If you keep your assets close by, criminals could target you for robbery. Professional criminals such as blackhat hackers and gangsters choose to act without the protection of a democratic nation-state as opposed to most people who prefer to stay civilians, with legal protection for ownership, such as assets in a bank. Mainstream adoption of crypto-assets requires that ordinary people protect their wealth from theft and extortion without relying on the legal system. This is the primary challenge of wallet technology.

Naturally Secure

What can nature tell us about keeping our valuables secure? Some animal species that save food for the future have, under the pressure of natural selection, become smart enough or strong enough to protect it. Bees, the makers of honey, employ an army of kamikaze pilots. They answer force with force. There are also other approaches used by non-violent animals. For example, the western scrub-jay does *cache spacing* : hiding food in separate locations. It spreads risk like a responsible portfolio manager. A competitor could discover one cache, but the other caches stay hidden. Squirrels and other rodents do *scatter hoarding* . They remember the exact location of about 1200 nuts and seeds. With similar “security through obscurity” measures, civilians could hold more wealth with less risk. Memory aids and hidden or protected backups are already part of the current best wallet designs.

Design Tradeoffs

Owning crypto-assets is, in some ways, easier than owning material things. You can transport your wealth anywhere safely, at little cost. Also, a private key is just words or numbers, making it easier to hide. However, the owner of a crypto-asset has to take more steps to deter coercion. Potential robbers have to be convinced that the owner can not access the keys now, or ever, under duress. Crypto-assets are the possessions of the future, but that future will not happen for mainstream users if holding those assets puts people at risk of violent robbery and extortion.

Chapter 1 mentioned the balance of constraints in *security* , *convenience* , and *freedom* (Dan Geer, *Cybersecurity and National Policy* , 2010). The first angle for innovation in wallet technology is leveling up tradeoffs

between convenience and security. The central dilemma is between *easy* and *risky* , or *complicated* and *safe* .

Any new, better wallet technology generation must transcend the limiting tradeoff of the previous generation user experience. A growing portion of non-technical people can keep their crypto-assets secure at a lower cost, with greater ease. Wallet designers aim to rise in that upward spiral of security and convenience. We can compare wallet innovation to the development of mobile phones into smartphones. The functionality and convenience expand with each new generation. Much of the wallet market will likely be seized by companies that combine hardware and software.

The tradeoff between convenience and security has leveled up three times already. We can watch the pattern and see how wallet technology is developing:

Exchange websites: The easiest way to buy and store your crypto-assets is to keep them on an exchange website. It is as easy as interacting with an online bank or brokerage account, but they only provide the illusion of security. The interface looks like a bank's, and your bank is mostly secure. However, the exchange website has the crypto-assets in custody, making for a 'honey pot' that attracts hackers. Crypto-assets are much easier to hack from an exchange than digital fiat currency from a bank. That risk is not theoretical. Exchanges get robbed regularly; there is a high-profile case in the news every few months. This primitive level of the tradeoff between convenience and security can not continue for long.

Hot wallets: The current crypto-wallets force the user into unpleasant choices. You can use 'hot wallets,' which stay connected to the internet, for receiving smaller amounts of crypto-assets that remain available for quick spending. The best hot wallets are well-designed and easy to use after the user understands the basics. The usability tempts the user to keep all their assets in that wallet, which is too risky. Carrying too much value in a hot wallet is not advisable because hackers can infect it remotely with a virus that steals your assets. Or robbers can take it by threatening the user, just like sometimes happens with valuable stores in a safe.

Hardware wallets: You can pay for more security and still retain some convenience with hardware wallets. You can always spend the assets without too much work, but it is more hassle than sending them from an exchange or hot wallet. Hardware wallets protect crypto assets better than exchange websites and hot wallets because it is harder to attack a hardware wallet remotely.

Multi-signature wallets and off-site backups: Some wallet solutions automatically keep a secure, off-site backup for recovery. Others can boost security by spreading the ability to spend over multiple locations, devices, or people.

Custodial cold wallets: For now, the highest combination of security and convenience is ‘cold storage’ with a custodian: someone else takes responsibility for securing your assets, and they keep the private keys offline, protected from remote hackers. You outsource the risk of theft and extortion, but you make yourself vulnerable to that third party. In case they are a company, they are regulated and insured like banks, and your assets are on the books for taxation. It also means that you accept the risk of the local authorities confiscating your crypto-assets, for whatever reason. In Geer’s triangle terms, you sacrifice a measure of freedom for a better deal on convenience and security.

In summary: Hardware wallets have leveled up the dilemma of *easy* versus *safe*. This is what the market has converged on for most situations. An even higher security level is possible with cold storage, where you expend more effort to keep your private keys in a physical form only without a connection to the internet. Cold storage with a custodian means that someone else holds the keys secure; you depend on a third party. For example, no one can get to your assets because they sit in a bunker in Switzerland.

Wallets are a multi-disciplinary design challenge covering hardware manufacturing, software security, “threat models,” mental models, and interface design. Companies that want to build a competitive wallet need a team that touches on all those fields. It can be a challenging journey. As a wallet customer, I have prepaid for a next-generation hardware wallet in a crowdfunding campaign, and their progress has halted along the way. I’m

rooting for them to finish and deliver a cutting-edge product, but it seems that wallets' market success is won the hard way. A company's rewards for designing and building wallets right are as high as the risks for getting it wrong.

Remove the Need for Custodians

A custodian who keeps your crypto-assets in cold storage is the equivalent of holding valuables in a safe deposit box with a traditional bank. You choose to trust them fully and minimize the risk of losing the keys through any error of your own. An adversary would need a court order to confiscate your crypto-assets, which is possible but in most situations, unlikely. If you encrypt the key, they cannot access your assets. Crypto-asset custodians come in two varieties: a) they are regulated, know how much you own, and have insurance against heists. Or b) they are unregulated like "money guards," they are a private person you trust, like a neighbor or family member. These money guards are standard in the developing world. These two types of custodians work well for many people, but they do not make use of the unique properties of crypto-assets. You and your money guard are still vulnerable to extortion, and you have to trust them.

To transcend the tradeoff between convenience and security, developers and entrepreneurs have to apply their imagination to make a wallet that is as secure as a custodian, without making the user vulnerable to a third party or potential robbers. That kind of wallet will support both the convenience of online exchange and the security of cold storage. Ideally, we do not want our assets under any central control. By rising to the challenge of storing crypto-assets safely without custodians, we move forward on a trajectory to disrupt the central authority of money and finance.

For next-generation wallets, startups can aspire to that security level, with users in full control of their wealth, with only a simple physical device that keeps the backup phrase and as easy to use as physical cash. Theft and loss protections are built-in. When holders can control and secure their crypto-assets without trusted third parties, banks and regulators will no longer have the power to frustrate owners according to random dictates. Typical current banking relationships have us paying third parties to handle everything, while customer experience is mediocre at best.

If we demand a smooth cryptocurrency experience, we have to bypass the banking system, which means exchanges such as Coinbase, Kraken, and

Bitstamp are only temporary solutions. They are too centralized and bureaucratic because they have to comply with banking regulations. We need something that gets crypto-assets into people's hands naturally, which is how new owners will become invested in the future of money.

Furthermore, we have to design the next generation of crypto-assets in a way that annihilates the incentive to hack or coerce. It should be impossible or prohibitively hard for potential thieves to reach crypto-assets. Owning crypto-assets must be safe and convenient for everyone so that all those independent owners together become a political constituency.

CHAPTER 6

Tokenized Assets

AS COMMUNICATION AND PAYMENTS BECOME IDEAL, frictionless transactions could change how people trade various assets. A popular blockchain can anchor transactions of assets other than money. Bitcoin and Ethereum are the current best candidates for such blockchains and would become a new and virtual jurisdiction. First, we will distinguish blockchain tokens from cryptocurrencies and other assets, and will view blockchain networks as platforms for the future of finance. After that, we look at the types of tokens that can live on these networks and what opportunities arise from them.

Creating a digital token representing a specific asset has an advantage: it expands the market for that asset, giving it greater liquidity. The owners of the tokenized asset can buy or sell it much faster than an ordinary investment. That by itself makes the asset more valuable. The token can represent a financial agreement, such as a debt, a physical property, gold or real estate. Porting the administration of such traditional assets over to a blockchain can make trade more efficient. However, the best chance of sparking opportunities comes from new types of assets, which cannot exist or be traded without digital tokens defined by a smart contract on a blockchain. For example, game-based digital items can interact and combine to make new and unique items, such as minting new digital trading cards or breeding creatures with individual digital genomes, which creators and owners can trade in a global marketplace.

Difference Between Currency and Tokens

As explained previously, the Ethereum network lets developers deploy smart contracts that define ETH flows on its blockchain or enable tokens. Many teams have launched these tokens to fund projects and to bootstrap a circular economy. To understand how these tokens work, we must first distinguish this type of crypto-asset from cryptocurrency. We call them “coins,” but that seems an attempt to link them with the success of Bitcoin. The word *coin* refers to a currency that circulates widely. Traditional,

physical coins do not represent anything else. They have a value of their own because they are scarce and valuable, either because of:

- the precious metal they are made of;
- the issuing central bank's precious metal;
- the fiat rights of the central bank.

Cryptocurrencies mimic the qualities of coins and paper cash. In contrast, tokens are not like cash but resemble laundromat or train ride tokens, or casino chips. They are portable objects that have value in a specific context. A token on a blockchain is a cryptographically secured digital asset that entitles the holder to something. Usually, it is a promise or guarantee made by the issuer. A token's value may be set by the issuer, as with casino chips, or imposed by the market, like physical commodities or collectible items. Even if the market determines the price, you might still need the issuer's technology, such as a website, to interpret the token. For example, imagine you are holding a digital cat token, a *CryptoKitty*, in your wallet, which you can only see or breed by connecting your wallet to the *CryptoKitties* -website. That means the ownership is decentralized, but not necessarily the asset's utility.

A cryptocurrency has value because the units, analogous to physical coins, are scarce *and* circulate widely. Both these qualities are united in virtual objects, the currency units on the blockchain. As described in previous chapters, these properties can catalyze a network effect. The units become a currency as increasing amounts of people use it.

In contrast, tokenized assets are scarce because tokens themselves are scarce or even unique, while their value depends on the asset they represent. The liquidity and scarcity of tokens depend on the network they reside in, but to appraise them, we also need to estimate the underlying asset's value.

Since the launch of altcoins and tokenized assets, some have tried to classify these new crypto-assets. Investors, speculators, and entrepreneurs need such a classification system to understand what they are buying or guide the design and issue of a token. In this chapter, we will only look at crypto-assets from the perspective of an entrepreneur who wants to create their own, for a) funding a project and b) bootstrapping a circular economy.

Blockchain Security Dilution

Bitcoin is the new standard for ideal money, with settlement assurances orders of magnitude more robust than any other monetary good (Nic Carter, *It's the settlement assurances, stupid*, 2019). It can serve as an anchor for smart contracts denominated in Bitcoin, including international transfers, payment rails, savings, contractual agreements, and derivative financial products. For the first time in history, these markets are global and neutral, with BTC's value as a single yardstick. It is universal in the same sense that people now calculate global prices in USD, and it is neutral in the spirit of gold: the value of—and control over—gold can hardly be influenced by a single actor, at least in the long term. Opportunities to build these financial functions and products will grow with software development that will expand on the Bitcoin network. We discussed the possibilities of Lightning Network in chapter 4.

Ethereum's goals from the outset have been different from those of Bitcoin. Although their vision has changed over time, Ethereum founders have always tried to build a system with more possibilities and lower security level. Early critics often mentioned the drawbacks of “a larger attack surface.” At the moment, critics deride ad-hoc improvisations to technical problems and fickle goals for the future. It seems that these criticisms are valid yet do not refute the central point: Ethereum is different enough from Bitcoin to blaze a trail with experiments, and is successful enough as measured by market capitalization. It is an open-source blockchain that keeps many people interested because a few times a year, a team launches something radically new on Ethereum. These novelties include the ERC-20 standard for tokens, the Decentralized Autonomous Organization (DAO), CryptoKitties, MakerDAO, bonding curves, etc. These novelties tend to multiply but can get in trouble. They can fail, have a crisis, or collapse, causing a loss for the risk-takers who bought the related assets. These problems lead to criticism, yet Ethereum has a track record as a platform for launching new crypto-assets to an existing user base: those who already own and use ETH. Any competing blockchain will have to overcome that network effect.

Bitcoin critics who focus on the cost of proof-of-work mining believe in new altcoins with a different consensus algorithm. The most common variant is proof-of-stake, which awards the block reward according to the amount of currency the miner, or “block producer,” has locked-up into a smart contract. Instead of ‘betting’ electricity on their block's validity,

block producers bet this staked amount on their commitment to producing honest blocks. Alternatively, a system can include voting and reputations for ‘delegated proof-of-stake.’ However, these types of algorithms are vulnerable to a political balance of power. Network participants want cheaper transactions on the blockchain, but drawback is a need for governance. To limit the cost of the network, developers have to dilute the safety of transactions.

Introducing governance to a blockchain seems strange to many Bitcoiners because they believe in transcending the need for governance through proof-of-work consensus. Governance tends to centralize. Bitcoiners accept the drawbacks of proof-of-work because they value decentralization, as an aid to immutability and censorship resistance. To them, decentralization is more important than the efficiency of transactions.

It seems that consensus algorithms that can replace proof-of-work mining are an open question for the blockchain industry. Future research and security breaches will reveal if governance-based asset networks will win the trust of the market. We have to see who wants to transact digital assets that are *not quite* unstoppable. Will people dare to transact pirate business on Ethereum without trouble?

Ethereum as a Laboratory

Although Ethereum is more centralized than Bitcoin, it might be decentralized *enough*. Its founders are betting on an organizational structure with identifiable people. It is a natural path to issuing a token that represents a claim to an asset. On the other hand, detractors see this organizational structure as evidence that the founders of Ethereum do not understand Bitcoin in the first place. They expect Ethereum and its organizations to fold under pressure from adversaries with the power of a nation-state. What will they do when influential politicians call their Switzerland office, demanding a ‘hard fork’ to shut down an ICO crowdfunding his or her opponents?

Regardless of the Ethereum blockchain’s long-term security issues, many exciting projects have found a home there. The success of some of these projects, and their tokens, attract new buyers of ETH and more developers to the ecosystem. So far, this network effect keeps Ethereum alive and growing.

Developers continue to build secondary protocols for Bitcoin, such as the Lightning Network and sidechains, expanding Bitcoin's inherent capabilities. We can imagine how the best applications of Ethereum smart contracts could also work with Bitcoin. If bitcoins are a universal payment method, and Ethereum's best smart contract applications enter the mainstream, some of these could succeed in the Bitcoin network.

For now, we have no way of predicting the future success of Ethereum versus Bitcoin. As an entrepreneur, you can keep up with the developments in both ecosystems until you spot an excellent opportunity. While you choose a product or service to build, know that you are tying the company's future to the network you build on.

Tokenized Fundraising

Since the ICO boom, which ran from 2016 to early 2018, we know that one use of smart contracts is fundraising. Instead of getting money from a bank, angel investor, or venture capital firm, you can source investment capital from anyone, anywhere, as long they have a wallet with crypto-assets. That expands the potential playing field of investment and funding a hundred times over. The amount of wealth that *could* exist as investable cryptocurrencies in crypto wallets *could* exceed the amount of money under the control of professional investors. Also, tokenized investing without anyone's permission speeds up the funding of projects, while investors can cash out faster. Many prefer investments that triple in value in six months to ones that grow ten times in value but take five years.

The most straightforward type of funding is issuing *security tokens*. These resemble traditional shares, entitling the holder to a portion of the company's profits. Shareholders are registered in government authority databases that oversee financial markets, while banks and brokers intermediate the trading of these shares for their clients. Security tokens disintermediate these third parties. A company can create a token, say that they represent a share in the company's profit, and promise to pay out these profits to any token holders. The blockchain maintains the record of ownership, instead of a central authority. Cryptography secures transfer between owners, instead of go-betweens.

The open playing field for issuing tokens comes with a new risk: new token buyers are often not protected by the regulations of conventional financial markets. That attracts teams who are better at selling than building

things, which includes outright scammers. Optimists maintain that the market will have to learn these lessons, while pessimists expect these dynamics to be the downfall of tokenized assets. No matter the outcomes of tokenized funding in coming years, the genie is out of the bottle.

Regulators will have to battle entrepreneurs who organize funding opportunities from competing jurisdictions. When ordinary people, often called “retail investors,” get a taste of outsized returns on investment, they will want to keep accessing these opportunities. Meanwhile, fundraisers will learn how to align incentives across stakeholders and mitigate the risk of scams, market manipulations, and unsophisticated investors. In traditional finance, these regulations developed around the turn of the 20th century, with the rise of retail investors.

Some crypto-professionals claim that security tokens misapply blockchain technology. Shares divide a corporation into pieces and entitle the shareholder to voting rights and profit. This type of organization is hundreds of years old, just like writing letters and contracts with ink on paper. Launching a security token resembles the post office launching a website that publishes its address, phone number, and opening hours while failing to prepare for a future in which email replaces most of their service. The only situation I view as fitting for a securities token is pseudonymous ownership of a traditional corporation. When a financial regulator allows the trade of pseudonymous tokens within their jurisdiction, it makes stock ownership possible for anyone, including people without a brokerage account. That still looks quite traditional. I think that crypto-innovators should aspire to more. For instance, the funding of brick-and-mortar companies can improve with tokenized shares or products. You can crowdfund on a global scale because you can tokenize your future products, which are then tradable. For example, the owners of traditional small businesses, such as restaurants, often need bank loans. With the launch of crypto-assets, they can instead sell prepaid meals to defer the risk to the customer. By giving a significant discount to future customers, they could disintermediate the bank.

Moreover, you no longer need to put up collateral. This decentralization of small business finance allows a broader range of people to start a restaurant or other brick-and-mortar companies. Another example: let’s say you wanted to start a software company in a developing nation. You can get

funding from anybody. What Kickstarter.com did for consumer products, tokenized assets can do for technology companies and service companies.

Designing Circular Economies

The section above shows how tokens can benefit the founders of issuing organizations. Entrepreneurs can get funding for their projects, but that does not in itself make tokens valuable. Tokens have to add value for other stakeholders in the project, or fail as an economic instrument. Tokens add friction to the experience of the product, therefore entrepreneurs who put a product behind a token ‘gate’ have to justify that choice. Why should people install a wallet for crypto-assets, go to an exchange to buy the token, or sign up on your website to receive them? They will only do so if the token gives them access to otherwise unavailable assets.

The first generation of tokenized assets show the challenge of designing a token that circulates well. These markets often have two sides: the platform for trading serves both buyers and sellers. The issuer opens a marketplace for a tradable resource, for example computing power (Golem) or data storage (Filecoin). The plan is for suppliers to get paid in the coin, and to sell it for fiat currency or cryptocurrency to pay their suppliers or to take profit. The token circulates between these parties as payment for resources. It also allows anyone to hold the traded supply in reserve, use it later, or rebalance when needed.

An asset representing access to a network or platform will have prices that follow the benefits and value of that network. As the number of people who use that network grows along with the system’s usefulness, demand rises. Token holders can sell their tokens for a profit, instead of using their tokens to access the network.

This token-mediated exchange of resources can cause a conflict of interest between stakeholders. The issuer chooses the number of tokens in circulation, while the utility, i.e., the benefit to the users, depends on the resource’s demand. The economics are described in the ‘Benefit and Scarcity Matrix’ (Sehra, Smith & Gomes, *The Economics of Initial Coin Offerings*, Allen & Overy, 2017). This type of token can end up in four positions:

- **Highly available, but not so useful:** challenging to circulate, unstable price.

- **Scarce, but not so useful:** mainly held by speculators, regulators could view it as a security.
- **Scarce, and useful:** leads to hoarding and deflation, causing friction in the market for the tradable resource.
- **Highly available and useful:** stable price, but requires a tangible product or service with known demand.

Some projects resolved these tensions between stakeholders by issuing their tokens slowly, adjusting the release to marketplace demand. The role they play in their token economy is similar to that of a fiat currency central bank; the token holders have to trust them to manage the supply and serve the interest of all the stakeholders. While that seemed to be an innocent responsibility with the first generation of digital assets, centralized management is against this technology's spirit, so we can expect future projects to issue tokens with distributed control.

Token designers are experimenting with ways to give stakeholders a proportionate voice or control in a project. Entrepreneurial teams are 'failing their way forward' to the right models for token governance, such as with the Ethereum Decentralized Autonomous Organization (DAO), proof-of-stake models, and MakerDAO. A standard to live up to is the decentralized, yet stable operation of the Bitcoin network. Bitcoin proves that stakeholders can maintain a delicate balance of power within a system of incentives, without any party having the final say. The question that other projects are exploring is, in essence, how to infuse other systems with the same qualities.

Once a blockchain network is widespread, none of the participants, whether they are traders, holders, miners, or developers, can force changes to the system against others participants' interests. If any of these groups try to change the rules, but the others stick with the software that applies the original rules, that means the rule-changers will only have a smaller, isolated network. In the jargon of open source software, this is called a 'fork.' The network effect ensures that a forked blockchain network is only as valuable as the number of people joining it.

A new project with a new model for keeping a balance of power gets scrutinized on its degree of decentralization—To what degree can its open transactions resist authoritarian censorship? As Bitcoin or Ethereum inspire

you, and you build something new, how will you make it live up to the standard set by these systems?

Game-Theoretical Balance

A decentralized network depends on a careful balance of interests between the decision-makers inside of a financial ecosystem. The strategic interactions of these decision-makers is described by a branch of mathematics called *game theory*. The designers of the novel Bitcoin-inspired systems tweak the candidate network parameters, aiming to support transactions. Besides the technical challenges related to cryptography and computer science, the design must foster relationships between mostly pseudonymous decision-makers. Each type of network participant needs reasons to a) play by the rules and b) enforce the rules for others. Game theory informs the design of a stable blockchain network.

The balance of rules and interests is critical; the founders of any project either borrow a proven model or attempt to invent a new one. So far, Ethereum is home to most of these experiments. In 2020, a branch of these experiments called ‘decentralized finance (DeFi)’ shows promise. It started with creating pseudonymous and transparent versions of ordinary banking products, such as savings and loans. The first decentralized stablecoin, Dai, results from a strategic balance between the holders of multiple tokens. As I write this, some teams are launching financial products that have not existed before, and cannot exist, within the methods of traditional finance.

The stablecoin Dai deserves special mention because it supports many other products in decentralized finance. A stablecoin, or crypto-dollar, which is entirely transparent in supply and function, is a must-have for the current adoption of crypto-assets. Later on, crypto-dollars could become obsolete, but as it is, the players in this ecosystem need assets to protect them from unwanted volatility. Furthermore, Dai is a case study in achieving strategic balance in the sense of game theory. As an entrepreneur launching a token, you benefit from knowing how that balance works. An entrepreneur’s education starts with Bitcoin, continues with Ethereum, and advances with MakerDAO’s Dai stablecoin. You can read an overview of the functioning of Dai in the MakerDAO whitepaper.

Non-Fungible Tokens

Besides currency and tokens, another class of crypto-assets relies on *non-fungible tokens* . If Bitcoin is ‘money with a blockchain,’ then a non-fungible token is ‘a collectible with a blockchain.’ “Fungible” means that one unit of the token is interchangeable with any other. If you receive a payment in dollars, bitcoin, or ETH, you don’t care which specific units you receive, when they were minted, and where they have been before. A coin is a coin, just as with fiat currency and casino chips. In contrast, a non-fungible token that the owner can send from one blockchain address to another is *not* interchangeable with any other token. It represents or refers to a unique asset. Examples of assets that could become non-fungible tokens are:

- Individual, secure, but tradable digital identities.
- Title deeds of real estate, vehicles, and works of art.
- Digital collectibles such as digital art or rare items inside of a game or virtual reality.
- Digital tickets for events and travel.

Non-fungible tokens introduce a new dimension of scarcity to the supply and demand of an asset. In the case of crypto-currency, its blockchain guarantees the rules that govern the issuance and transfer of the token. These public and verifiable rules ensure the scarcity of each of the units of the token. A non-fungible token relies on its blockchain to provide an additional guarantee: the token is X-of-a kind. There is a limited number of units of this token in circulation, where for most use cases, it is unique and refers to a single asset.

The applications on non-fungible tokens I listed above are all promising. However, they each have unresolved issues. I am excited about the individual attempts of some existing projects, and I recommend these areas as sources of entrepreneurial opportunity.

Besides financial applications, the area that is showing the most promise is online games. Gaming startups that figure out how to make online gaming more enjoyable for players can tap into a vast market.

Digital Collectibles in Games

Non-fungible tokens were first applied successfully in a game by *AxiomZen* , as part of the game *CryptoKitties* , launched in 2017. The goal was to

collect, breed, and trade cartoon cats. The cats have genes that interact and can express specific traits. The challenge for the players is to figure out how a trait depends on which genes to breed rare cats. You can sell a cat for ETH, or you can rent it out for breeding. Some breeders have made quite a lot of ETH playing this game. Other players must have lost money if they invested in a collection while the market was peaking or failed to breed the kitties with profit.

The technical progress behind the *CryptoKitties* -game is that each cat is unique, with genes defined by a smart contract, and ‘lives’ as a token at the owner’s address on the Ethereum blockchain. The game engine interprets the token and renders the smart contract’s computer code as a visible cat on the game’s website. When a player breeds two cats, a new token is minted by combining and updating the tokens referring to its parent tokens. The rules introduce scarcity in two ways: First, when a cat breeds, there is a cooling-off period before it can breed again. Each time it breeds, this period increases so that the cat becomes a slower breeder. Second, later generations breed slower too. The most valuable cats have rare traits, are first-generation, and have bred little or not at all. Cats with normal genes, of later generations, and much offspring are so common that no one wants to buy them. They sit at the player’s Ethereum address and game profile, just being cute.

The company behind *CryptoKitties* used the features of the Ethereum blockchain to move computer gaming forward. They needed a robust platform for smart contracts, with a native and widely traded token—like ETH. Players no longer depend on the gaming company to steward their in-game assets, particularly, the cats. As tokens at an Ethereum address, players have full control over them. Transactions and breeding results are immutable. Although they still rely on the central game engine, which runs on the company’s servers, anyone could build a game engine that interprets the tokens and renders them as cartoon cats on a website.

Representing in-game items as tokens gives rise to another possibility: developing a new game that repurposes pre-existing digital assets. For example, *KittyRace* lets you race any of your *CryptoKitties* against those of other players. You bet ETH on your kitty, and the winner takes the pot. These game mechanics led to some players breeding cats with features that made them good at winning races: a feedback loop between the two games about the same set of in-game assets. Another project is *KittyHats* , to outfit

your cats with apparel and accessories. These projects suggest that a robust network effect will bring more players and developers, inspired to build new decentralized games that use the existing in-game assets.

CryptoKitties is a proof of concept for rare digital in-game assets. Before this game, people who handled cryptocurrencies and tokens were motivated by technical innovation, financial gain, or political activism. *CryptoKitties* proved that people would transact a crypto-asset because it was collectible; Kitties are not always assets to be sold; they can be valuable and exciting for their own sake. Players can enjoy breeding, collecting, and trading cats, without any interest in the technology that makes it possible. Even though most of the players were crypto-enthusiasts, and many play for profit, they don't have to. Just as *Atari*'s popular game *Pong* proved that computers could generate profits as entertainment platforms, *CryptoKitties* showed that blockchains can run for fun.

Since *CryptoKitties*, several games were launched that use a blockchain to make the game elements scarce. For example, *Gods Unchained* is a strategy game with rare cards, which players can combine into even rarer cards. Just as with tradable cats, not every player has the same reason for playing. Some want to buy rare cards to win matches, while others want to combine and collect cards to sell their tokens for profit. The marketplace for cards is part of the game world.

Another game in the making that uses digital scarcity is *BeachHead 2020*, a project that I was part of. *BeachHead 2020* is a first-person shooter in virtual reality that plans to grow into a full virtual reality economy with scarce digital goods, such as weapons, fashion, cars, and land. It has the aesthetic of cyberpunk warfare, and real estate in the city will be limited, just as it is in Hong Kong or Manhattan. Players will have to stake BeachHead coin to get the title deed to real estate. Designing the finer points of this economy was an exciting challenge for the team.

The use of a blockchain as an entertainment platform suggests new types of games with rare and tradable assets. The financial element blends the game world with the real world. Traditional online games stumbled into this phenomenon by accident. When massive multiplayer online role-playing games such as *World of Warcraft* became popular, a market grew for in-game items, in-game currency, and even whole characters. The virtual world *Second Life* also grew an economy of virtual goods. The

economics of *Fortnight* show how players are willing to pay for purely aesthetic additions to their in-game character; virtual fashion is real.

From the perspective of blockchain trades, these non-tokenized markets look clumsy. By having virtual items exist as tokens, they are much easier and safer to trade for currency. Furthermore, the possibility to take tokenized in-game assets into other games and virtual worlds means that game designers can catalyze feedback loops such as between *CryptoKitties* and *KittyRace* .

Building the Next Phase of the Ecosystem

Non-fungible tokens inside of online games are only one area poised for hypergrowth. In 2020, Decentralized Finance is heating up, with many projects launching volatile tokens. In previous years, Ethereum was hosting projects that made everyday financial products transparent, expanding on traditional financial transactions such as borrowing and lending with collateral. Projects use the Ethereum blockchain to create new financial products and techniques, such as bonding curves and flash loans. Beyond the heated speculation, teams on the cutting edge can imagine new financial instruments that traditional players cannot dream up. Virtual jurisdictions empower the fast growth of new kinds of blockchain assets built by entrepreneurs. Liberated from the permission of financial authorities such as banks, they create products involving money and contracts, without the limits imposed by arbitrary third parties. These new companies make up a growing ecosystem of stakeholders, enjoying opportunities in a wide-open field. Eventually, this can lead to Decentralized Autonomous Organizations (DAOs) that take over the market for laws and agreements. They can host contracts and regulations on a global scale, free from geographical jurisdictions. Say goodbye to the hassles of banks and go-betweens!

Conclusion

IN THIS BOOK'S INTRODUCTION, I pointed at the tired practices in global finance that are now ready for a transformation, or even revolution, by entrepreneurs who harness Bitcoin, Ethereum, and other blockchains. I claimed that innovation is predatory, and I invited you on an adventure. I promised this book would serve as your map to navigate the confusion on this frontier. Here at the end of the book, I hope you have found insight into the industry and inspiration to start your project.

Any financial function that makes one depend on a third party is a candidate for automation on a decentralized blockchain. A third party's presence creates a need for an immutable ledger of transactions. Most chapters include ideas for starting a company based on cryptocurrency and tokenized assets. You now know the attack angles on the legacy system by applying Bitcoin and blockchains to create better financial products. The better the entrepreneur understands the past and present of money and trade, the better they will envision and build its future.

The territory of freedom is a developing ecosystem, where the first types of companies to survive and thrive are speculators, traders, and market makers. In Chapter 3, I emphasized liquidity and the catalysts of network effects. This established layer of speculation and trade serves as a base as the next layer matures, including payment systems, smart contracts, and wallets.

Then we explored these younger types of companies, who thrive in virtual jurisdictions on popular blockchains. Lightning Network makes the 'good old Bitcoins' more spendable, and DeFi is an experimental lab that competes with traditional finance. If ETH2.0 lives up to its promises, we can expect efficient decentralized applications at scale, available for mainstream users. Did any of these chapters give you an idea for a project? I look forward to hearing about your plans.

In the coming years, we might get a sense of what a Bitcoin-based financial system looks like. Near the end of 2020, the global economy is depressed because of the Covid-19 pandemic, and a growing part of the population sees through the magic trick of fiat money. The political and

business elites discuss “The Great Reset” of the economy and financial system at the World Economic Forum. Meanwhile, a growing number of prominent investors and executives have become Bitcoin enthusiasts and affirm its value in public. These milestones come together as an excellent foundation for another bull market. When the prices of Bitcoin and other crypto-assets take off, the time is right for seeking investors and customers for your crypto-venture.

As long as the Bitcoin network does not yet handle a large number of smaller transactions, BTC stays a commodity akin to gold: individuals and institutions can use it to store and transfer large sums of money. If—when—Bitcoin *does* scale, with sidechains and Lightning Network, it will completely revive “hard” commodity money, with the added benefit of frictionless and private transactions. Developers and entrepreneurs will continue to invent new financial assets and operations, building toward the future of money that is secure, convenient, and free of central control.

I hope you will join me in being a part of the future.

About the Author

STEFAN KING HAS STUDIED BITCOIN since 2013 and became an active evangelist in 2016. He has a track record in software marketing and applied himself to the intricacies of launching crypto-assets. As a host of the Weekly Bitcoin Meetup in Chiang Mai, he explains Bitcoin to beginners and advises entrepreneurs building on Bitcoin and Ethereum. You can find him at www.stefantking.com and at twitter.com/stefantking .